

[Home](#) > CCNA – New Questions Part 10

CCNA – New Questions Part 10

June 6th, 2022 [Go to comments](#)

Premium Members: You can practice these questions first via these links:

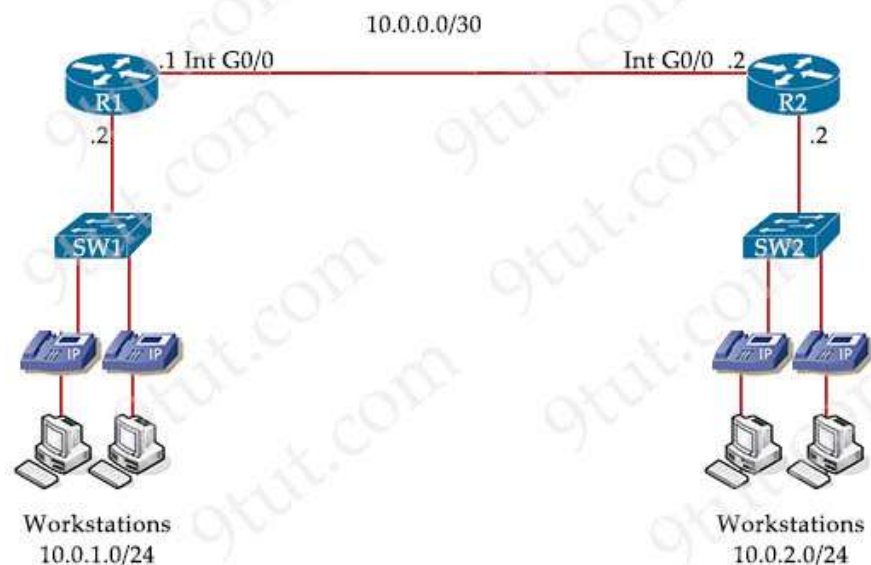
- + [Question 1 to 20](#)
- + [Question 21 to 39](#)
- + [Question 40 to 57](#)
- + [Question 58 to 78](#)
- + [Question 79 to 95](#)

Or practice all 95 questions of this part 10 at:

- + [All CCNA Questions – Part 10](#) (all questions in one page; shuffled)
- + [All CCNA Questions – Part 10 Practice Mode](#) (one question per page, unshuffled)

Question 1

Refer to the exhibit.



An engineer is asked to configure router R1 so that it forms an OSPF single-area neighbor relationship with R2. Which command sequence must be implemented to configure the router?

- A. router ospf 10
network 10.0.0.0 0.0.0.3 area 0
network 10.0.2.0 0.0.0.255 area 0
- B. router ospf 10
network 10.0.0.0 0.0.0.3 area 0
network 10.0.1.0 0.0.0.255 area 0
- C. router ospf 10
network 10.0.0.0 0.0.0.3 area 0
network 10.0.2.0 255.255.255.0 area 0
- D. router ospf 10
network 10.0.0.0 0.0.0.252 area 0
network 10.0.1.0 0.0.0.255 area 0

Answer: B

Question 2

Drag and drop the HTTP methods used with REST-Based APIs from the left onto the descriptions on the right.

GET	creates a resource and returns to URI in the response header
DELETE	creates or replaces a previously modified resource using information in the request body
PATCH	removes a resource
PUT	retrieves a list of a resource's URIs
POST	updates a resource using instructions included in the request body

Answer:

- + creates a resource and returns to URI in the response header: POST
- + creates or replaces a previously modified resource using information in the request body: PUT
- + removes a resource: DELETE
- + retrieves a list of a resource's URIs: GET
- + updates a resource using instructions included in the request body: PATCH

Explanation

GET: retrieve data

POST: create data

PUT: fully update (i.e. replace) an existing record

PATCH: update part of an existing record

DELETE: delete records

PUT is similar to POST in that it can create resources, but it does so when there is a defined URL wherein PUT replaces the entire resource if it exists or creates new if it does not exist.

Unlike PUT Request, PATCH does partial update. Fields that need to be updated by the client, only that field is updated without modifying the other field.

Question 3

What is the collapsed layer in collapsed core architectures?

- A. core and WAN
- B. access and WAN
- C. distribution and access
- D. core and distribution

Answer: D

Explanation

Collapsed core networks can save the company money by reducing the amount of hardware needed to build and run the network. By combining the core and distribution layers, this model eliminates the need for separate devices. This offers companies the chance to utilize most of the benefits offered by the three-tiered model in a budget-friendly option.

Question 4

What is the MAC address used with VRRP as a virtual address?

- A. 00-00-0C-07-AD-89
- B. 00-00-5E-00-01-0a
- C. 00-07-C0-70-AB-01
- D. 00-C6-41-93-90-91

Answer: B

Explanation

With VRRP, the virtual router's MAC address is 0000.5E00.01xx , in which xx is the VRRP group.

Note: We are not sure why there is a VRRP question here as it is not in the syllabus of the exam.

Question 5

Drag and drop the IPv6 addresses from the left onto the corresponding address types on the right.

fcba:926a:e8e:7a25:b1:c6d2:1a76:8fdc	Global Unicast
2001:db8:600d:cafe::123	Link-Local Unicast
fe80:a00:27ff:feeb:8eaa	Multicast
ff05::1:3	Unique Local

Answer:

Global unicast: 2001:db8:600d:cafe::123

Link-Local unicast: fe80:a00:27ff:feeb:8eaa

Multicast: ff05::1:3

Unique Local: fcba:926a:e8e:7a25:b1:c6d2:1a76:8fdc

Explanation

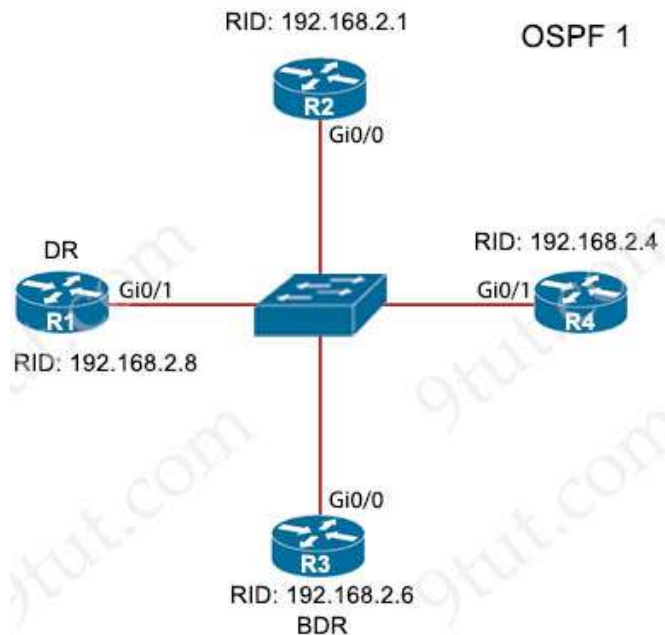
Link-local addresses are allocated with the FE80::/64 prefix -> can be easily recognized by the prefix FE80

All IPv6 **multicast** addresses share the prefix of FF00::/8 -> The first octet is FF (1111 1111). This way you can tell at a glance if an IPv6 address is intended for multicast or not.

Unique local IPv6 addresses have the similar function as IPv4 private addresses. They are not allocated by an address registry and are not meant to be routed outside their domain. Unique local IPv6 addresses start with FC00::/7 -> The first octet can be FC or FD. However when you implement this you have to set the L-bit (the right-most bit of the first octet) to 1 which means that the first two digits will be FD.

Question 6

Refer to the exhibit.



All routers in the network are configured. R2 must be the DR. After the engineer connected the devices, R1 was elected as the DR. Which command sequence must be configured on R2 to be elected as the DR in the network?

- A. R2(config)#interface gi0/0
R2(config-if)#ip ospf priority 1
- B. R2(config)#interface gi0/0
R2(config-if)#ip ospf priority 100
- C. R2(config)#router ospf 1
R2(config-router)#router-id 10.100.100.100
- D. R2(config)#router ospf 1
R2(config-router)#router-id 192.168.2.7

Answer: B

Question 7

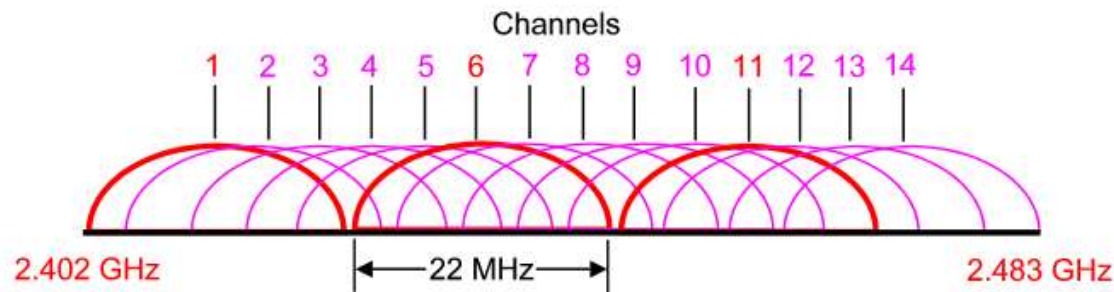
Which set of 2.4 GHz nonoverlapping wireless channels is standard in the United States?

- A. channels 2, 7, 9, and 11
- B. channels 1, 6, 11, and 14
- C. channels 2, 7, and 11
- D. channels 1, 6, and 11

Answer: D

Explanation

The 2.4 GHz band is subdivided into multiple channels each allotted 22 MHz bandwidth and separated from the next channel by 5 MHz.
-> A best practice for 802.11b/g/n WLANs requiring multiple APs is to use non-overlapping channels such as 1, 6, and 11.



Question 8

Which command entered on a switch configured with Rapid-PVST+ listens and learns for a specific time period?

- A. switch(config)#spanning-tree vlan 1 max-age 6
- B. switch(config)#spanning-tree vlan 1 hello-time 10
- C. switch(config)#spanning-tree vlan 1 priority 4096
- D. switch(config)#spanning-tree vlan 1 forward-time 20

Answer: D

Explanation

The command spanning-tree vlan 1 forward-time ...” controls how fast a port changes its spanning tree state from blocking to forwarding.

===== New Questions (added on 16th-Jun-2022) =====

Question 9

Drag and drop the Wi-Fi terms from the left onto the descriptions on the right.

extended service set	Wi-Fi option based around one or more access points
distribution system	Wi-Fi option in which cells from different access points are linked together
infrastructure mode	alphanumeric text string that identifies a wireless network
SSID	Wi-Fi option that enables two or more clients to communicate directly without a central access point
independent basic service set	entire wireless cell of an access point and the linkage to the wired network

Answer:

- + Wi-Fi option based around one or more access points: distribution system
- + Wi-Fi option in which cells from different access points are linked together: extended service set
- + alphanumeric text string that identifies a wireless network: SSID
- + Wi-Fi option that enables two or more clients to communicate directly without a central access point: independent basic service set
- + entire wireless cell of an access point and the linkage to the wired network: infrastructure mode

Explanation

A Wireless **Distribution System** (WDS) lets you connect multiple access points together. WDS allows the connected access points to communicate with each other via wireless connection. This feature enables clients who roam to have a seamless experience. This makes it easier to manage multiple wireless networks as well as reduces the amount of cables required to connect the networks.

Reference: <https://www.cisco.com/c/en/us/support/docs/smb/wireless/cisco-small-business-100-series-wireless-access-points/smb2040-connect-multiple-access-points-together-through-wireless-dis.html>

Access Points are limited devices. They can cover a Basic Service Area (BSA) and it is called a Cell. But in many companies network area is very large that only one Access Point can not cover all the area. So, additional Access Points are used to cover all the area. This Access Points are connected through an Ethernet Switch and they become a single continuous and seamless connection from the client perspective. User can go from one cell to another without connection interrupt. So, combining one more Access Point in a switched network is called **Extended Service Set** (ESS).

Reference: <https://ipccisco.com/lesson/wireless-principles/>

Independent Basic Service Set (IBSS), as the name suggests, is a service set that allows wireless devices or stations to communicate with each other without any medium or central device. It does not contain or include any AP (Access Point). That's why it is also known as an ad hoc mode or peer to peer network.

Extended Service Set (ESS): uses more than one AP to create a WLAN, allows roaming in a larger area than a single AP.

Infrastructure mode is an 802.11 networking framework in which devices communicate with each other by first going through an Access Point (AP). In infrastructure mode, wireless devices can communicate with each other or can communicate with a wired network.

Question 10

Refer to the exhibit.

```
interface g2/0/0
 channel-group 1 mode active
interface g4/0/0
 channel-group 1 mode active
interface Port-channel1
 ip address 203.0.113.65 255.255.255.252
```

%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to down

An engineer is configuring a Layer 3 port-channel interface with LACP. The configuration on the first device is complete, and it is verified that both interfaces have registered the neighbor device in the CDP table. Which task on the neighbor device enables the new port channel to come up without negotiating the channel?

- A. Bring up the neighboring interfaces using the **no shutdown** command.
- B. Change the EtherChannel mode on the neighboring interfaces to **auto**
- C. Modify the static EtherChannel configuration of the device to **passive** mode
- D. Configure the IP address of the neighboring device

Answer: C

Explanation

Our switch can see neighbor switch in the CDP table which means the interfaces of both ends have been turned on (with no shutdown command) -> Answer A is not correct.

Our switch is using LACP (active/passive mode) so we cannot use PAgP (auto/desirable mode) on the neighboring interface -> Answer B is not correct.

The error is at Layer 2 (“changed state to down”) so it is not about “configuring the IP address” of the neighboring device which is at Layer 3 -> Answer D is not correct.

Therefore only answer C is left. We have to change the Etherchannel mode of neighboring device from static to passive for LACP.

Question 11

Refer to the exhibit.

R1#show ip route

Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2

ia - IS-IS inter area, * - candidate default, U - per-user static route

o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/24 is subnetted, 5 subnets

D 10.1.2.0/24 [90/2170112] via 10.145.20.226, 00:01:33, Serial0/0

D 10.1.3.0/24 [90/2170112] via 10.145.20.226, 00:01:33, Serial0/0

D 10.1.2.0/25 [90/2170112] via 10.145.20.126, 00:01:33, Serial0/0

D 10.1.3.0/25 [90/2170112] via 10.145.20.146, 00:01:33, Serial0/0

D 10.1.4.0/25 [90/2170112] via 10.145.20.156, 00:01:33, Serial0/0

192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks

C 192.168.10.0/24 is directly connected, GigabitEthernet0/0

192.168.21.0/24 is variably subnetted, 2 subnets, 2 masks

C 192.168.11.0/24 is directly connected, GigabitEthernet0/1

10.165.20.0/24 is variably subnetted, 2 subnets, 2 masks

C 192.168.20.224/24 is directly connected, Serial0/0

S 10.1.2.112/28 [1/0] via 10.165.20.166

What is the next hop for traffic entering R1 with a destination of 10.1.2.126?

A. 10.165.20.126

B. 10.165.20.146

C. 10.165.20.166

D. 10.165.20.226

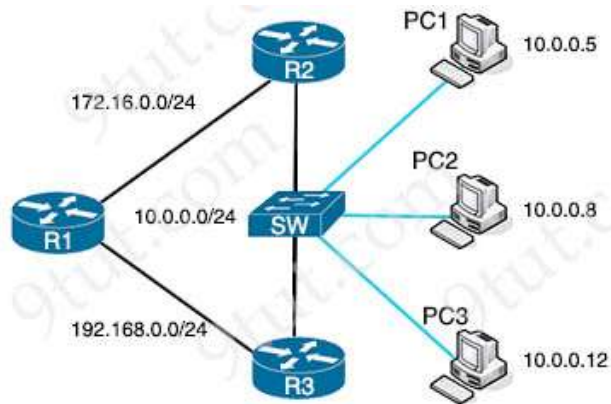
Answer: C

Explanation

The static route destined to subnet 10.1.2.112/28 which ranges from 10.1.2.112 to 10.1.2.127 (increment: 16) so it also covers the destination of 10.1.2.126. The static route also has the lowest AD (of 1) so the router will choose this path.

Question 12

Refer to the exhibit.



A network engineer must configure R1 so that it sends all packets destined to the 10.0.0.0/24 network to R3, and all packets destined to PC1 to R2. Which configuration must the engineer implement?

- A. R1(config)#ip route 10.0.0.0 255.255.0.0 172.16.0.2
R1(config)#ip route 10.0.0.5 255.255.255.255 192.168.0.2
- B. R1(config)#ip route 10.0.0.0 255.255.255.0 172.16.0.2
R1(config)#ip route 10.0.0.5 255.255.255.255 192.168.0.2
- C. R1(config)#ip route 10.0.0.0 255.255.0.0 192.168.0.2
R1(config)#ip route 10.0.0.0 255.255.255.0 172.16.0.2
- D. R1(config)#ip route 10.0.0.0 255.255.255.0 192.168.0.2
R1(config)#ip route 10.0.0.5 255.255.255.255 172.16.0.2

Answer: D

Question 13

Drag and drop the facts about wireless architectures from the left onto the types of access point on the right. Not all options are used.

configured and managed by a WLC	Autonomous Access Point
accessible for management via Telnet SSH, or a Web GUI	
supports different operational modes	
requires a management IP address	Lightweight Access Point
supports automatic deployment	

Answer:

Autonomous Access Point

- + accessible for management via Telnet SSH, or a Web GUI
- + requires a management IP address

Lightweight Access Point

- + configured and managed by a WLC
- + supports different operational modes

Explanation

An autonomous AP is a self-contained device with both wired and wireless hardware so that it can bridge to the wired VLAN infrastructure wireless clients that belong to SSIDs. Each autonomous AP must be configured with a management IP address so that it can be remotely accessed using Telnet, SSH, or a web interface. Each AP must be individually managed and maintained unless you use a management platform such as Cisco DNA Center.

Reference: <https://www.ciscopress.com/articles/article.asp?p=2999384&seqNum=5>

Question 14

Drag and drop the functions of SNMP fault-management from the left onto the definitions on the right.

fault detection	The network management system launches a preconfigured script to restore functionality
event correlation and aggregation	The administrator can manually intervene at the source of the fault
problem resolution	The system identifies performance degradation or service interruption
restoration of service	The system groups alarms from related issues
fault diagnosis and isolation	The system reports on the source of the issue

Answer:

- + The network management system launches a preconfigured script to restore functionality: restoration of service
- + The administrator can manually intervene at the source of the fault: problem resolution
- + The system identifies performance degradation or service interruption: fault detection
- + The system groups alarms from related issues: event correlation and aggregation
- + The system reports on the source of the issue: fault diagnosis and isolation

Explanation

Fault management is the component of network management concerned with detecting, isolating and resolving problems. Properly implemented, network fault management can keep connectivity, applications and services running at an optimum level, provide fault tolerance and minimize downtime. Platforms or tools designed specifically for this purpose are called fault management systems.

- + **Fault detection:** The system discovers that service delivery has been interrupted or its performance has degraded.
- + **Fault diagnosis and isolation:** The source of the fault, such as a component failure or power outage, and its location in the network topology are identified.
- + **Event correlation and aggregation:** A single fault can cause multiple alarms, fault management systems often group related events for administrators and provide a root cause analysis.
- + **Restoration of service:** The network management system automatically executes any preconfigured scripts or programs to get services up and running as soon as possible.
- + **Problem resolution:** The source of the fault is corrected, repaired or replaced. Depending on the cause, manual intervention may be required.

Reference: <https://www.manageengine.com/network-monitoring/network-fault-remediation.html>

Question 15

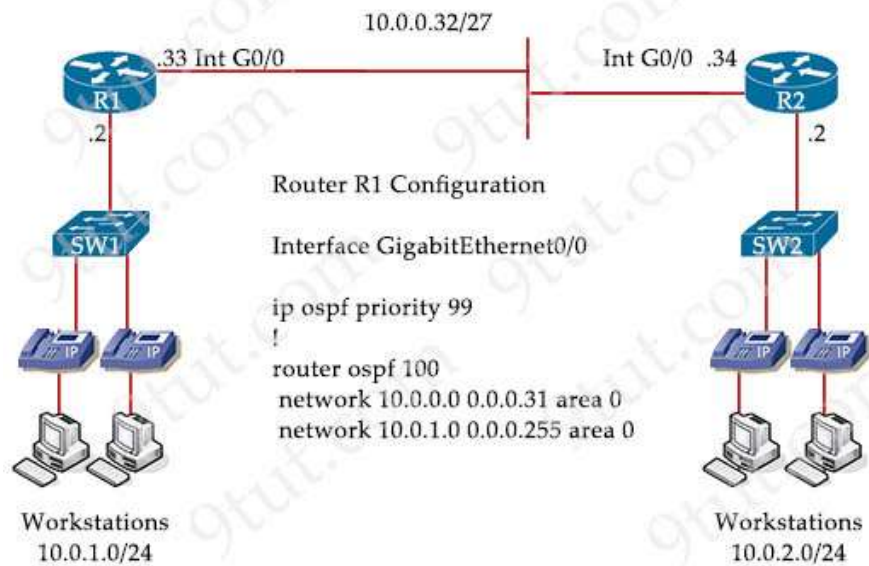
What is the purpose of the Cisco DNA Center controller?

- A. to securely manage and deploy network devices
- B. to scan a network and generate a layer 2 network diagram
- C. to provide Layer 3 services to autonomous access points
- D. to secure physical access to a data center

Answer: A

Question 16

Refer to the exhibit.



An engineer must configure router R2 so it is elected as the DR on the WAN subnet. Which command sequence must be configured?

- A. interface gigabitethernet0/0
ip address 10.0.1.1 255.255.255.224
ip ospf priority 98
- B. interface gigabitethernet0/0
ip address 10.0.1.1 255.255.255.0
ip ospf priority 255
- C. interface gigabitethernet0/0
ip address 10.0.0.34 255.255.255.248
ip ospf priority 0
- D. interface gigabitethernet0/0
ip address 10.0.0.34 255.255.255.224
ip ospf priority 100

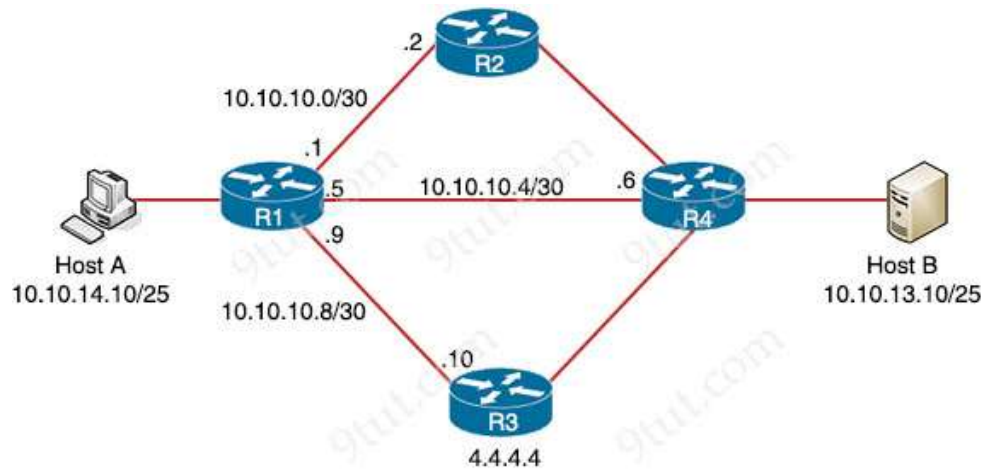
Answer: D

Explanation

The OSPF priority of R1 is 99 so we have to set the OSPF priority to a higher value. Also the IP address must be 10.0.0.34/27.

Question 17

Refer to the exhibit.



```
R1(config)#ip route 0.0.0.0 0.0.0.0 10.10.10.2
R1(config)#ip route 10.10.13.0 255.255.255.128 10.10.10.2 111
R1(config)#ip route 10.10.13.0 255.255.255.128 10.10.10.6 112
R1(config)#ip route 10.10.13.0 255.255.255.128 10.10.10.10 108
R1(config-router)#router ospf 1
R1(config-router)#router-id 1.1.1.1
R1(config-router)#network 10.10.10.5 0.0.0.0 area 0
R1(config-router)#network 10.10.10.1 0.0.0.0 area 0
R1(config-router)#network 10.10.14.1 0.0.0.0 area 0
```

R1 has just received a packet from host A that is destined to host B. Which route in the routing table is used by R1 to reach host B?

- A. 10.10.13.0/25[1/0] via 10.10.10.2
- B. 10.10.13.0/25[110/2] via 10.10.10.6
- C. 10.10.13.0/25[110/2] via 10.10.10.2
- D. 10.10.13.0/25[108/0] via 10.10.10.10

Answer: D

Explanation

The “10.10.13.0/25[1/0] via 10.10.10.2” entry is from default route (ip route 0.0.0.0 0.0.0.0 10.10.10.2) so it is only used when no other routes to 10.10.13.0/25 is found.

In three static routes, the best match for host B should be “ip route 10.10.13.0 255.255.255.128 10.10.10.10 108” with lowest AD of 108 so R1 would choose this path. This AD is also lower than the AD of OSPF so it is preferred over OSPF learned routes.

Question 18

Refer to the exhibit.

```
C:\Users\ciscoadmin>ipconfig/all
Windows IP Configuration

Host Name . . . . . : DESKTOP-480JBET
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
DNS Suffix Search List. . . . . : arcap.se

Ethernet adapter Ethernet:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Realtek PCIe GBE Family
Controller
Physical Address. . . . . : 3C-52-82-33-F3-8F
DHCP Enabled . . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . : arcap.se
Description . . . . . : Intel(R) Dual Band
Wireless-AC 7265
Physical Address. . . . . : C821-5B-84-F3-EF
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::45a1:b3fa:2f37:bf37%2 (Preferred)
IPv4 Address. . . . . : 192.168.1.226 (Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : October 3, 2019 12:28:08 PM
Lease Expired . . . . . : October 3, 2019 7:18:38 PM
Default Gateway . . . . . : 192.168.1.100
DHCP Server . . . . . : 192.168.1.254
DHCPv6 IAID . . . . . : 4667016B
DHCPv6 Client DUID. . . . . : 00-01-00-01-20-FF-05-55-3C-F3-34-29-20-DF
DNS Servers . . . . . : 192.168.1.253
NetBIOS over Tcpip. . . . . : Enabled
Connection-specific DNS Suffix Search List:
    arcap.se
```

The given Windows PC is requesting the IP address of the host at www.cisco.com. To which IP address is the request sent?

- A. 192.168.1.226
- B. 192.168.1.253
- C. 192.168.1.100
- D. 192.168.1.254

Answer: B

Explanation

In this question, the Windows PC is trying to resolve the domain name `www.cisco.com` to an IP address so it will need a DNS server.

From the output we see the line “DNS Servers ...: 192.168.1.253” so this is the IP address the request is sent. Notice that the “Default Gateway” IP address is in the same subnet of DNS Server so the PC can directly send request to the DNS Server.

Question 19

What is a feature of TFTP?

- A. provides secure data transfer
- B. relies on the well-known TCP port 20 to transmit data
- C. uses two separate connections for control and data traffic
- D. offers anonymous user login ability

Answer: D

Explanation

TFTP is a simple protocol for transferring files, implemented on top of the UDP/IP protocols using well-known port number 69 -> Answer A and answer B are not correct.

TFTP has no session control -> Answer C is not correct.

Question 20

Which access point mode relies on a centralized controller for management, roaming, and SSID configuration?

- A. repeater mode
- B. bridge mode
- C. lightweight mode
- D. autonomous mode

Answer: C

Explanation

Lightweight: The term ‘lightweight’ refers to the fact that these devices cannot work independently. A Cisco lightweight AP (LAP) has to join a Wireless LAN Controller (WLC) to function.

Question 21

Which command creates a static NAT binding for a PC address of 10.1.1.1 to the public routable address 209.165.200.225 assigned to the PC?

- A. R1(config)#ip nat outside source static 209.165.200.225 10.1.1.1
- B. R1(config)#ip nat inside source static 209.165.200.225 10.1.1.1
- C. R1(config)#ip nat outside source static 10.1.1.1 209.165.200.225
- D. R1(config)#ip nat inside source static 10.1.1.1 209.165.200.225

Answer: D

Question 22

Refer to the exhibit.

```
RIP    10.1.1.16/28[120/5]  via F0/0
OSPF   10.1.1.0/24[110/30]  via F0/1
OSPF   10.1.1.0/24[110/40]  via F0/2
EIGRP  10.1.0.0/26[90/20]   via F0/3
EIGRP  10.0.0.0/8[90/133]   via F0/4
```

Packets received by the router from BGP enter via a serial interface at 209.165.201.1. Each route is present within the routing table. Which interface is used to forward traffic with a destination IP of 10.1.1.19?

- A. F0/0
- B. F0/1
- C. F0/3
- D. F0/4

Answer: A

Question 23

Which two REST API status-code classes represent errors? (Choose two)

- A. 1XX
- B. 2XX
- C. 3XX
- D. 4XX
- E. 5XX

Answer: D E

Explanation

HTTP defines these standard status codes that can be used to convey the results of a client's request. The status codes are divided into five categories.

1xx: Informational – Communicates transfer protocol-level information.

2xx: Success – Indicates that the client's request was accepted successfully.

3xx: Redirection – Indicates that the client must take some additional action in order to complete their request.

4xx: Client Error – This category of error status codes points the finger at clients.

5xx: Server Error – The server takes responsibility for these error status codes.

Question 24

An engineer has configured the domain name, user name, and password on the local router. What is the next step to complete the configuration for a Secure Shell access RSA key?

- A. crypto key generate rsa
- B. crypto key pubkey-chain rsa
- C. crypto key import rsa pem
- D. crypto key zeroize rsa

Answer: A

Explanation

Steps to configure SSH:

1. Configure the router hostname using command "hostname".
2. Configure the domain name using command "ip domain-name".
3. Generate public and private keys using command "crypto key generate rsa".
4. Create a user in the local database using command "username...secret".
5. Allow only SSH access on VTY lines using command "transport input ssh".

Reference: <https://ipwithease.com/how-to-configure-ssh-version-2-on-cisco-router/>

Question 25

Which encryption method is used by WPA3?

- A. TKIP
- B. SAE
- C. PSK
- D. AES

Answer: D

Explanation

From the below figure, we can see that SAE is only the authentication method. WPA3 uses AES for encryption method both for WPA3-Personal:

The image shows the Cisco WLAN configuration interface for a WLAN named 'Secure Me'. The 'Security' tab is selected, and the 'Personal' security type is chosen. The 'WPA2+WPA3 Parameters' section shows 'WPA3' selected and 'CCMP128(AES)' as the encryption cipher. The 'Fast Transition' section shows 'Disable' selected. The 'Protected Management Frame' section shows 'Required' selected. The 'Authentication Key Management' section shows 'ASCII' selected for PSK Format and 'Enable' checked for SAE.

WLANs

WLANs

Advanced

WLANs > Edit 'Secure Me'

General **Security** **QoS** **Policy-Mapping** **Advanced**

Security Type: Personal

MAC Filtering: ☐

WPA2+WPA3 Parameters

Policy: ☐ WPA2 ☒ WPA3

Encryption Cipher: ☒ CCMP128(AES)

Fast Transition

Fast Transition: Disable

Protected Management Frame

PMF: Required

Comeback timer(1-10sec): 1

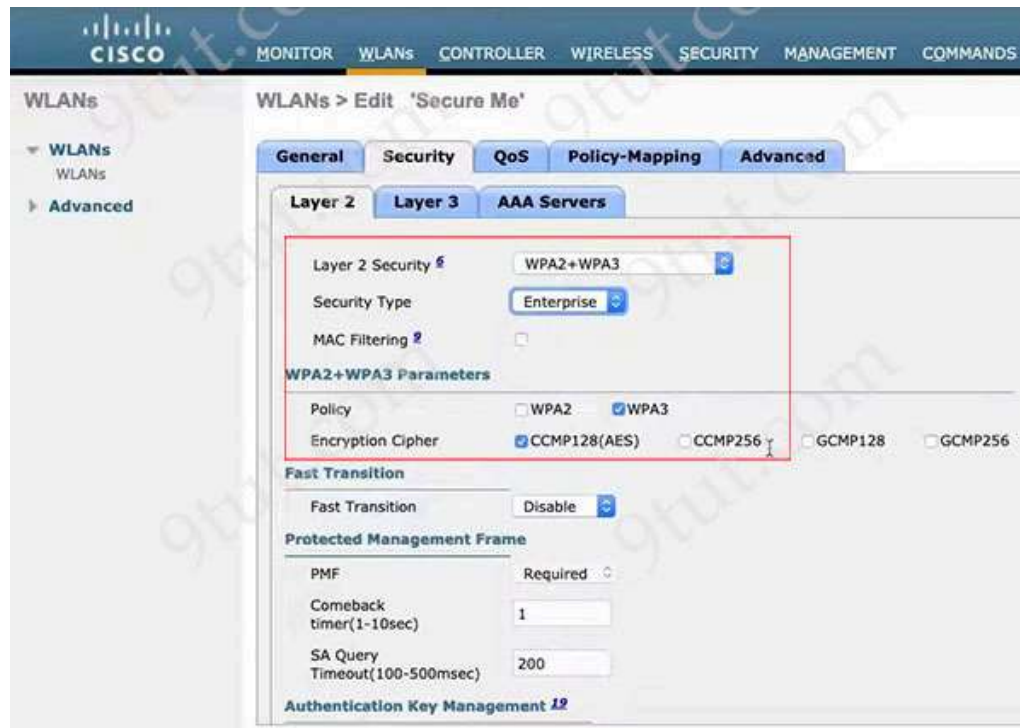
SA Query Timeout(100-500msec): 200

Authentication Key Management

PSK Format: ASCII

SAE: ☒ Enable

and WPA3-Enterprise:



===== New Questions (added on 13th-Aug-2022) =====

Question 26

An engineer is configuring remote access to a router from IP subnet 10.139.58.0/28. The domain name, crypto keys, and SSH have been configured. Which configuration enables the traffic on the destination router?

A. interface FastEthernet0/0

ip address 10.122.49.1 255.255.255.248

ip access-group 10 in

ip access-list standard 10

permit udp 10.139.58.0 0.0.0.7 host 10.122.49.1 eq 22

B. interface FastEthernet0/0

ip address 10.122.49.1 255.255.255.252

ip access-group 105 in

ip access-list standard 105

permit tcp 10.139.58.0 0.0.0.7 eq 22 host 10.122.49.1

C. interface FastEthernet0/0

ip address 10.122.49.1 255.255.255.252

ip access-group 110 in

```
ip access-list extended 110
permit tcp 10.139.58.0 0.0.0.15 host 10.122.49.1 eq 22
```

```
D. interface FastEthernet0/0
ip address 10.122.49.1 255.255.255.240
access-group 120 in
```

```
ip access-list extended 120
permit tcp 10.139.58.0 255.255.255.248 any eq 22
```

Answer: C

Question 27

What is a function of spine-and-leaf architecture?

- A. mitigates oversubscription by adding a layer of leaf switches
- B. limits payload size of traffic within the leaf layer
- C. offers predictable latency of the traffic path between end devices
- D. exclusively sends multicast traffic between servers that are directly connected to the spine

Answer: C

Explanation

A spine-leaf architecture aids this by ensuring traffic is always the same number of hops from its next destination, so latency is lower and predictable.

Reference: <https://www.arubanetworks.com/faq/what-is-spine-leaf-architecture/>

Question 28

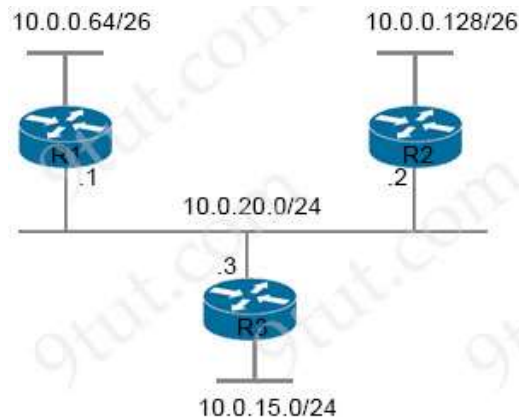
What differentiates device management enabled by Cisco DNA Center from traditional campus device management?

- A. CLI-oriented device
- B. centralized
- C. per-device
- D. device-by-device hands-on

Answer: B

Question 29

Refer to the exhibit.



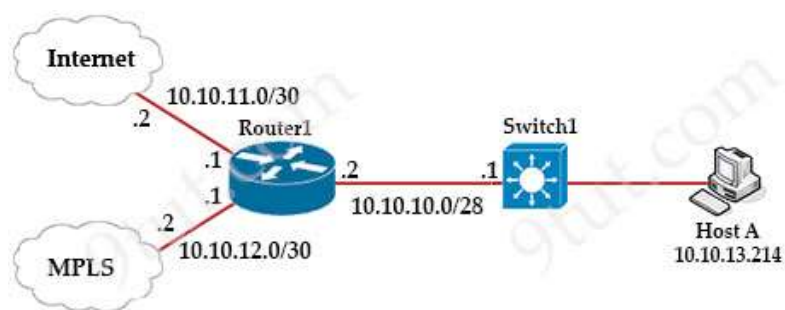
Router R1 is added to the network and configured with the 10.0.0.64/26 and 10.0.20.0/24 subnets. However, traffic destined for the LAN on R3 is not accessible. Which command when executed on R1 defines a static route to reach the R3 LAN?

- A. ip route 10.0.15.0 255.255.255.0 10.0.20.1
- B. ip route 10.0.15.0 255.255.255.192 10.0.20.1
- C. ip route 10.0.0.64 255.255.255.192 10.0.20.3
- D. ip route 10.0.15.0 255.255.255.0 10.0.20.3

Answer: D

Question 30

Refer to the exhibit.



Router1#show ip route
Gateway of last resort is 10.10.11.2 to network 0.0.0.0

- 209.165.200.0/27 is subnetted, 1 subnets
- B 209.165.200.224 [20/0] via 10.10.12.2, 03:03:03
- 209.165.201.0/27 is subnetted, 1 subnets
- B 209.165.201.0 [20/0] via 10.10.12.2, 03:03:03
- 209.165.202.0/27 is subnetted, 1 subnets

```

B    209.165.202.128 [20/0] via 10.10.12.2, 03:03:03
    10.0.0.0/8 is variably subnetted, 8 subnets, 4 masks
C    10.10.10.0/28 is directly connected, GigabitEthernet0/0
C    10.10.11.0/30 is directly connected, FastEthernet2/0
C    10.10.12.0/30 is directly connected, GigabitEthernet0/1
O    10.10.13.0/25 [110/2] via 10.10.10.1, 00:00:03, GigabitEthernet0/0
O    10.10.13.128/28 [110/2] via 10.10.10.1, 00:00:03, GigabitEthernet0/0
O    10.10.13.144/28 [110/2] via 10.10.10.1, 00:00:03, GigabitEthernet0/0
O    10.10.13.160/29 [110/2] via 10.10.10.1, 00:00:03, GigabitEthernet0/0
O    10.10.13.208/29 [110/2] via 10.10.10.1, 00:00:03, GigabitEthernet0/0
S*   0.0.0.0/0 [1/0] via 10.10.11.2

```

Drag and drop the prefix lengths from the left onto the corresponding prefixes on the right. Not all prefixes are used.

255.255.255.128	10.10.13.0
255.255.255.224	10.10.13.144
255.255.255.240	10.10.13.160
255.255.255.248	209.165.202.128
255.255.255.252	

Answer:

10.10.13.0 – 255.255.255.128
 10.10.13.144 – 255.255.255.240
 10.10.13.160 – 255.255.255.248
 209.165.202.128 – 255.255.255.224

Question 31

What is a zero-day exploit?

- A. It is when an attacker inserts malicious code into a SQL server.
- B. It is when a new network vulnerability is discovered before a fix is available.
- C. It is when the perpetrator inserts itself in a conversation between two parties and captures or alters data.
- D. It is when the network is saturated with malicious traffic that overloads resources and bandwidth.

Answer: B

Question 32

After a recent security breach and a RADIUS failure, an engineer must secure the console port of each enterprise router with a local username and password. Which configuration must the engineer apply to accomplish this task?

Option A <pre>aaa new-model aaa authorization exec default local aaa authentication login default radius username localuser privilege 15 secret plaintextpassword</pre>	Option B <pre>username localuser secret plaintextpassword line con 0 login authentication default privilege level 15</pre>
Option C <pre>username localuser secret plaintextpassword line con 0 no login local privilege level 15</pre>	Option D <pre>aaa new-model line con 0 password plaintextpassword privilege level 15</pre>

- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer: B (?)

Explanation

Option A is not correct as in fact the command “aaa authentication login default” should be “aaa authentication login default **group** radius” which will authenticate only via RADIUS server.

Option C is not correct as the command “no login local” removes the requirement for local authentication on the console line.

Option D is not correct as with the command “password plaintextpassword”, the device only asks for the password, not username.

Option B is not correct as the command “login authentication default” specifies the default login authentication method but we need to define that “default” method is to use local username and password with two additional commands “aaa new-model” and “aaa authentication login default local”. -> This is the best choice although it is not totally correct.

===== New Questions (added on 26th-Aug-2022) =====

Question 33

Which command implies the use of SNMPv3?

- A. snmp-server community
- B. snmp-server host
- C. snmp-server enable traps
- D. snmp-server user

Answer: D

Explanation

Only SNMPv3 requires to add a SNMP user to a group to work.

Note: “community” string is the password which is used in SNMPv1 & v2. “snmp-server host” command can be used by both SNMPv2 & v3.

Question 34

Refer to the exhibit.

Layer 2 **Layer 3** **AAA Servers**

Layer 2 Security **WPA+WPA2**

MAC Filtering ☐

Fast Transition

Fast Transition **Adaptive**

Over the DS ☒

Reassociation Timeout **20** Seconds

Protected Management Frame

PMF **Disabled**

WPA+WPA2 Parameters

WPA Policy ☐

WPA2 Policy ☒

WPA2 Encryption ☒ AES ☐ TKIP ☐ CCMP256 ☐ GCMP128 ☐ GCMP256

OSN Policy ☐

Authentication Key Management

802.1X ☒ Enable

CCKM ☐ Enable

PSK ☐ Enable

FT 802.1X ☐ Enable

FT PSK ☐ Enable

SUITEB-1X ☐ Enable

SUITEB192-1X ☐ Enable

WPA gtk-randomize State **Disable**

Clients on the WLAN are required to use 802.11r. What action must be taken to meet the requirement?

- A. Enable CCKM under Authentication Key Management
- B. Under Protected Management Frames, set the PMF option to Required
- C. Set the Fast Transition option to Enable and enable FT 802.1X under Authentication Key Management
- D. Set the Fast Transition option and the WPA gtk-randomize State to disable

Answer: C

Explanation

Configuring 802.11r Fast Transition (GUI)

Procedure

Step 1 Choose WLANs to open the WLANs window.

Step 2 Click a WLAN ID to open the WLANs > Edit window.

Step 3 Choose Security > Layer 2 tab.

Step 4 From the Layer 2 Security drop-down list, choose WPA+WPA2.

The Authentication Key Management parameters for Fast Transition are displayed.

Step 5 From the Fast Transition drop-down list, choose Fast Transition on the WLAN.

Step 6 Check or uncheck the Over the DS check box to enable or disable Fast Transition over a distributed system.

This option is available only if you enable Fast Transition or if Fast Transition is adaptive.

To use 802.11r Fast Transition over-the-air and over-the-ds must be disabled.

Step 7 In the Reassociation Timeout field, enter the number of seconds after which the reassociation attempt of a client to an AP should time out. The valid range is 1 to 100 seconds.

Note: This option is available only if you enable Fast Transition.

Step 8 Under Authentication Key Management, choose FT 802.1X or FT PSK.

Reference: <https://www.cisco.com/c/dam/en/us/td/docs/wireless/controller/technotes/80211r-ft/b-80211r-dg.html>

Fast BSS Transition (802.11r, often abbreviated to Fast Transition or FT) describes mechanisms by which a mobile device can reestablish existing security and/or QoS parameters prior to reassociating to a new AP. These mechanisms are referred to as “fast” because they seek to significantly reduce the length of time that connectivity is interrupted between a mobile device and Wi-Fi infrastructure when that mobile device is connecting to a new AP. Please note that the process of disconnecting from one AP and connecting to another AP is formally designated as a “BSS transition”. Therefore, the protocols established by FT apply to mobile device transitions between APs only within the same mobility domain and within the same ESS (ESS transition is out of scope for FT)

Reference: <https://blogs.cisco.com/networking/what-is-802-11r-why-is-this-important>

===== New Questions (added on 26th-Sep-2022) =====

Question 35

When a switch receives a frame for an unknown destination MAC address, how is the frame handled?

- A. broadcast to all ports on the switch
- B. flooded to all ports except the origination port
- C. forwarded to the first available port
- D. inspected and dropped by the switch

Answer: B

Question 36

What is the default port-security behavior on a trunk link?

- A. It causes a network loop when a violation occurs.
- B. It disables the native VLAN configuration as soon as port security is enabled.
- C. It places the port in the err-disabled state if it learns more than one MAC address.
- D. It places the port in the err-disabled state after 10 MAC addresses are statically configured.

Answer: C

Explanation

We tested it with IOS v15.2 and this is the result:

```
Switch#sh run int e0/0
Building configuration...

Current configuration : 116 bytes
!
interface Ethernet0/0
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport port-security
end

Switch#sh port-security int e0/0
Port Security          : Enabled
Port Status            : Secure-up
Violation Mode         : Shutdown
Aging Time             : 0 mins
Aging Type             : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses  : 1
Total MAC Addresses    : 0
Configured MAC Addresses : 0
Sticky MAC Addresses   : 0
Last Source Address:Vlan : 0000.0000.0000:0
Security Violation Count : 0

Switch#
```

We can see the “Maximum MAC Addresses” is 1 so this port will be put in the err-disabled state if it learns more than one MAC address.

Question 37

Refer to the exhibit.

```
{
  "aaaUser": {
    "attributes": {
      "pwd": "password1",
      "firstName": "Abraham",
      "lastName": "Lincoln",
      "phone": "5555551212",
      "email": "test@cisco.com"
    },
    "children": [{
      "aaaUserRole": {
        "attributes": {
          "name": "ExampleCisco"
        },
        "children": [{
          "aaaUserRole": {
            "attributes": {
              "name": "admin"
            }
          }
        ]
      }
    ]
  }
}
```

How many objects are present in the given JSON-encoded data?

- A. one
- B. four
- C. seven
- D. nine

Answer: D

Explanation

Objects are enclosed in braces “{}” so we only need to count the number of “}” to specify how many objects in the JSON-encoded data. And there are 9 closing brackets “}” so there are 9 objects.

Question 38

What are two examples of multifactor authentication? (Choose two)

- A. single sign-on
- B. unique user knowledge
- C. passwords that expire

- D. soft tokens
- E. shared password responsibility

Answer: B D

Explanation

A multi-factor authentication example of something the user knows could include: Passwords. PIN (or personal identification numbers). Answers to supposedly secret questions (such as “Where were you born?” or “The name of your first-grade teacher.”)

Question 39

Refer to the exhibit.

Router1#show ip route

Gateway of last resort is 10.10.11.2 to network 0.0.0.0

```

    209.165.200.0/27 is subnetted, 1 subnets
B       209.165.200.224 [20/0] via 10.10.12.2, 03:22:14
    209.165.201.0/27 is subnetted, 1 subnets
B       209.165.201.0 [20/0] via 10.10.12.2, 02:26:33
    209.165.202.0/27 is subnetted, 1 subnets
B       209.165.202.128 [20/0] via 10.10.12.2, 02:26:03
    10.0.0.0/8 is variably subnetted, 8 subnets, 4 masks
C       10.10.10.0/28 is directly connected, GigabitEthernet0/0
C       10.10.11.0/30 is directly connected, FastEthernet2/0
C       10.10.12.0/30 is directly connected, GigabitEthernet0/1
O       10.10.13.0/25 [110/2] via 10.10.10.1, 00:00:04, GigabitEthernet0/0
O       10.10.13.128/28 [110/2] via 10.10.10.1, 00:00:04, GigabitEthernet0/0
O       10.10.13.144/28 [110/2] via 10.10.10.1, 00:00:04, GigabitEthernet0/0
O       10.10.13.160/29 [110/2] via 10.10.10.1, 00:00:04, GigabitEthernet0/0
O       10.10.13.208/29 [110/2] via 10.10.10.1, 00:00:04, GigabitEthernet0/0
S* 0.0.0.0/0 [1/0] via 10.10.11.2

```

What is the subnet mask of the route to the 10.10.13.160 prefix?

- A. 255.255.255.240
- B. 255.255.255.128
- C. 255.255.248.0
- D. 255.255.255.248

Answer: D

Explanation

The 10.10.13.160 prefix has subnet mask of /29 which is 255.255.255.248.

===== New Questions (added on 26th-Oct-2022) =====

Question 40

Refer to the exhibit.



Routers R1, R2, and R3 use a protocol to identify their neighbors' IP addresses, hardware platforms, and software versions. A network engineer must configure R2 to avoid sharing any neighbor information with R3, and maintain its relationship with R1. What action meets this requirement?

- A. Configure the **no lldp run** command globally
- B. Configure the **no lldp receive** command on g0/1
- C. Configure the **no cdp run** command globally
- D. Configure the **no cdp enable** command on g0/2

Answer: D

Explanation

In order to disable CDP on an interface, we have to use the “no cdp enable” under interface mode.

If we want to disable sending LLDP on g0/2 only then we have to use the command “no lldp transmit”.

Question 41

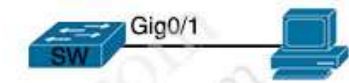
What is a function of an endpoint?

- A. It passes unicast communication between hosts in a network
- B. It is used directly by an individual user to access network services
- C. It provides security between trusted and untrusted sections of the network
- D. It transmits broadcast traffic between devices in the same VLAN

Answer: B

Question 42

Refer to the exhibit.



```

SW# conf t
SW(config)#interface gigabitEthernet0/1
SW(config-if)#switchport mode access
SW(config-if)#switchport port-security
SW(config-if)#
  
```

A network engineer started to configure port security on a new switch. These requirements must be met:

- MAC addresses must be learned dynamically.
- Log messages must be generated without disabling the interface when unwanted traffic is seen.

Which two commands must be configured to complete this task? (Choose two)

- A. SW(config-if)#switchport port-security mac-address 0010.7B84.45E6
- B. SW(config-if)#switchport port-security maximum 2
- C. SW(config-if)#switchport port-security mac-address sticky
- D. SW(config-if)#switchport port-security violation shutdown
- E. SW(config-if)#switchport port-security violation restrict

Answer: B E

Explanation

The requirement said MAC addresses must be learned dynamically so we cannot assign a specific MAC address or use the “sticky” keyword -> Answer A and answer C are not correct. Also the requirement said “MAC addresses” in plural so we set the maximum to 2 -> Answer B is correct.

“Log messages must be generated without disabling the interface” -> use “restrict” keyword for violation, not “shutdown” keyword -> Answer E is correct.

The “sticky” keyword does not mean the MAC addresses will be learned dynamically. It means the learned MAC addresses would be put into the running-config only. The question asks “MAC addresses must be learned dynamically” means we don’t configure static MAC learning only.

Note:

You can configure the interface for one of these violation modes, based on the action to be taken if a violation occurs:

- + **Restrict** – A port security violation restricts data, causes the SecurityViolation counter to increment, and causes an SNMP Notification to be generated.
- + **Shutdown** – A port security violation causes the interface to shut down immediately. When a secure port is in the error-disabled state, you can bring it out of this state by entering the errdisable recovery cause psecure_violation global configuration command or you can manually reenale it by entering the shutdown and no shut down interface configuration commands. This is the default mode.

Question 43

What are two features of the DHCP relay agent? (Choose two)

- A. minimizes the necessary number of DHCP servers
- B. assigns DNS locally and then forwards request to DHCP server

- C. is configured under the Layer 3 interface of a router on the client subnet
- D. allows only MAC-to-IP reservations to determine the local subnet of a client
- E. permits one IP helper command under an individual Layer 3 interface

Answer: A C

Explanation

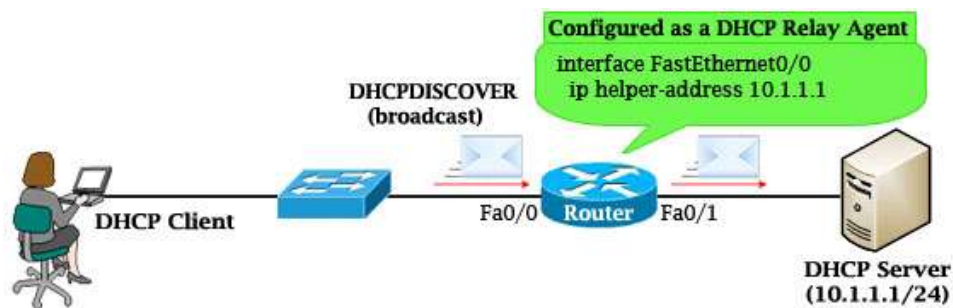
With the DHCP relay agent, we can configure DHCP servers in different subnet of the DHCP client. Otherwise we have to place one DHCP server in the same subnet of DHCP clients -> Answer A is correct.

Note: We can configure more than one ip helper command under one Layer 3 interface:

```
CE1#sh run int e0/0
Building configuration...

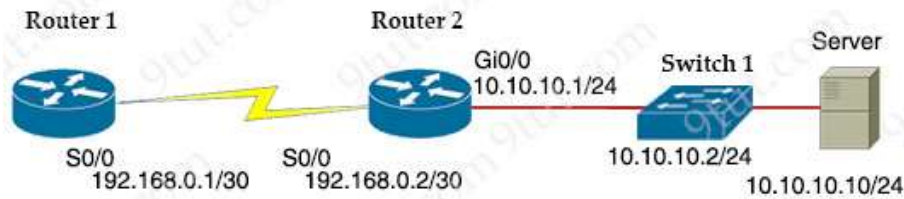
Current configuration : 123 bytes
!
interface Ethernet0/0
ip address 10.0.0.2 255.255.255.252
ip helper-address 10.1.1.1
ip helper-address 10.1.10.1
end
```

To make a router a DHCP Relay Agent, simply put the “ip helper-address <IP-address-of-DHCP-Server>” command under the interface that receives the DHCP messages from the DHCP Client -> Answer C is correct.



Question 44

Refer to the exhibit.



A network engineer must configure router R1 with a host route to the server. Which command must the engineer configure?

- A. R1(config)#ip route 10.10.10.10 255.255.255.255 192.168.0.2
- B. R1(config)#ip route 10.10.10.0 255.255.255.0 192.168.0.2
- C. R1(config)#ip route 0.0.0.0 0.0.0.0 192.168.0.2
- D. R1(config)#ip route 192.168.0.2 255.255.255.255 10.10.10.10

Answer: A

Explanation

A host route is an IPv4 address with a 32-bit mask (255.255.255.255).

Question 45

In a cloud-computing environment, what is rapid elasticity?

- A. automatic adjustment of capacity based on need
- B. control and monitoring of resource consumption by the tenant
- C. pooling resources in a multitenant model based on need
- D. self-service of computing resources by the tenant

Answer: A

Explanation

Rapid elasticity allows users to automatically request additional space in the cloud or other types of services. Because of the setup of cloud computing services, provisioning can be seamless for the client or user. The fact that providers still need to allocate and de-allocate resources is often irrelevant on the client or user's side. This is a very essential aspect of cloud technology. In a sense, cloud computing resources appear to be infinite or automatically available. That's much different from older systems, where the limits of storage or memory were immediately visible to a user.

Reference: <https://www.techopedia.com/definition/29526/rapid-elasticity>

Question 46

Drag and drop the steps in a standard DNS lookup operation from the left into the order on the right.

The DNS submits a request to the domain DNS server	Step 1
The DNS responds to the endpoint	Step 2
An endpoint submits a request for the IP address of a domain name	Step 3
The DNS submits a request to a root DNS server	Step 4
The DNS receives a reply from the domain DNS server	Step 5

Answer:

Step 1: An endpoint submits a request for the IP address of a domain name

Step 2: The DNS submits a request to a root DNS server

Step 3: The DNS submits a request to the domain DNS server

Step 4: The DNS receives a reply from the domain DNS server

Step 5: The DNS responds to the endpoint

Explanation

Maybe the author of this question wanted to hide the word “local” from the DNS so we should understand “The DNS” means “The local DNS Server”.

A DNS lookup is the process by which a host queries a domain name and a DNS server returns an IP address.

The eight steps in a DNS lookup:

1. A user enters a domain name (e.g., 9tut.com) into their browser, and the browser sends the query via their internet service provider ISP to a DNS recursive resolver.
2. The DNS recursive resolver, in turn, sends a query to the **root DNS nameserver**.
3. The root server returns to the resolver the address of the top-level domain (i.e., “TLD”) DNS root server, which has the needed information for the 9tut.com domain. (Examples of a top-level domain tld include “.com”, “.net”, and “.org,” which each TLD having it’s own root DNS server.)
4. In turn, the resolver then sends the information request to the Top-Level Domain server (In this case, the “.com” TLD nameserver).
5. The TLD name server responds to the resolver with the targeted IP address of the domain’s nameserver. (In this case, the DNS server for “9tut.com”).
6. Next, the DNS recursive resolver sends the query to the **domain’s DNS server**.
7. The domain’s DNS server then returns the IP address to the DNS resolver for the requested domain (e.g., “9tut.com”).
8. Finally, the DNS resolver returns the IP address of the requested domain to the requesting web browser. The browser sends the HTTPS request to the targeted IP address, and the server with that address returns the webpage, which renders in the user’s browser.

Reference: <https://www.infoblox.com/glossary/domain-name-system-dns/>

Question 47

What must be considered for a locally switched FlexConnect AP if the VLANs that are used by the AP and client access are different?

- A. The APs must be connected to the switch with multiple links in LAG mode.
- B. The native VLAN must match the management VLAN of the AP.
- C. The switch port mode must be set to trunk.
- D. IEEE 802.1Q trunking must be disabled on the switch port.

Answer: C

Explanation

FlexConnect VLANs and ACLs

You can configure the LAN uplink interface of a FlexConnect AP as either an access port or as a trunk. If you configure the interface as an access port, then the AP's management traffic and all client traffic, whether centrally or locally switched, will be in the same VLAN. For security and reliability reasons, we recommend that you segregate the client traffic from the management VLAN, and so to configure the AP's switchport as a trunk, with separately tagged VLANs for locally switched client traffic.

Reference: https://www.cisco.com/c/en/us/td/docs/wireless/controller/8-5/config-guide/b_cg85/flexconnect.html

Answer B is not correct as it should be “The native VLAN must match the native VLAN of the AP” (not the management VLAN).

Question 48

Which command configures the Cisco WLC to prevent a serial session with the WLC CLI from being automatically logged out?

- A. config sessions maxsessions 0
- B. config serial timeout 9600
- C. config serial timeout 0
- D. config sessions timeout 0

Answer: C

Explanation

The CLI automatically logs you out without saving any changes after 5 minutes of inactivity. You can set the automatic logout from 0 (never log out) to 160 minutes using the **config serial timeout** command.

To prevent SSH or Telnet sessions from timing out, run the **config sessions timeout 0** command.

Reference: https://www.cisco.com/c/en/us/td/docs/wireless/controller/8-0/configuration-guide/b_cg80/b_cg80_chapter_011.html

This question asks about the serial session so the first command is the correct answer.

Question 49

Which two IPv6 addresses are used to provide connectivity between two routers on a shared link? (Choose two)

- A. 2002::512:1204b:1111::1/64
- B. ff06:bb43:cc13:dd16:1bb:ff14:7545:234d

- C. FF02::0001:FF00:0000/104
- D. 2001:701:104b:1111::1/64
- E. ::ffff:10.14.101.1/96

Answer: D E (?)

Explanation

To provide connectivity between two routers on a shared link, **global unicast addresses** or **link-local addresses** are typically used. Answer D is correct as this is a global unicast address, valid for router-to-router communication on a shared link.

Answer A is not correct as it has two double colons "::". This answer also has another error in "1204b" which has 5 hexadecimal numbers in an octet (an IPv6 octet only has a maximum of 4 hexadecimal numbers). But in the exam please check this answer carefully. If its syntax is correct then this is the correct answer.

FF02::0001:FF00:0000 is an IPv6 multicast address that is part of the solicited-node multicast group (starts with FF02::1:FF/104). This group is automatically created for each configured IPv6 address on an interface. When a device needs to send an IPv6 packet to a specific device, it first sends a Neighbor Solicitation message to the solicited-node multicast address for the destination IPv6 address. The router receiving this message responds with a Neighbor Advertisement message containing its MAC address.

The address ff06:bb43:cc13:dd16:1bb:ff14:7545:234d is a site-local IPv6 multicast address.

Multicast addresses are used for group communication, not direct connectivity between two routers so answer B and answer C are not correct.

Note: An IPv6 Multicast address is always started with FF (in hex).

Note about ::ffff IPv6 address:

IPv4-mapped IPv6 address uses this alternative format. This type of address is used to represent IPv4 nodes as IPv6 addresses. It allows IPv6 applications to communicate directly with IPv4 applications. For example, 0:0:0:0:ffff:192.1.56.10 and ::ffff:192.1.56.10/96 (shortened format).

Reference: <https://www.ibm.com/docs/en/i/7.2?topic=concepts-ipv6-address-formats>

Question 50

Refer to the exhibit.

General **Security** **QoS** **Policy-Mapping** **Advanced**

Off Channel Scanning Defer

Scan Defer Priority: 0 1 2 3 4 5 6 7
☐ ☐ ☐ ☐ ☒ ☒ ☒ ☐

Scan Defer Time(msecs): 100

FlexConnect

FlexConnect Local Switching ☐ Enabled

FlexConnect Local Auth ☐ Enabled

Learn Client IP Address ☒ Enabled

Vlan based Central Switching ☐ Enabled

Central DHCP Processing ☐ Enabled

Override DNS ☐ Enabled

NAT-PAT ☐ Enabled

Central Assoc ☐ Enabled

Lync

Lync Server: Disabled

Local Client Profiling

HTTP Profiling ☐

DHCP Profiling ☐

HTTP Profiling ☐

PMIP

PMIP Mobility Type ☐

PMIP NAI Type: Hexadecimal

PMIP Profile: None

PMIP Realm:

Universal AP Admin Support

Universal AP Admin ☐

11v BSS Transition Support

BSS Transition ☒

Disassociation Imminent ☐

Disassociation Timer(0 to 3000 TBTT): 200

Optimized Roaming Disassociation Timer(0 to 40 TBTT): 40

An architect is managing a wireless network with APs from several branch offices connecting to the WLC in the data center. There is a new requirement for a single WLAN to process the client data traffic without sending it to the WLC. Which action must be taken to complete the request?

- A. Enable **local HTTP profiling**
- B. Enable **FlexConnect Local Switching**
- C. Enable **local DHCP Profiling**
- D. Enable **Disassociation Imminent**

Answer: B

Explanation

FlexConnect AP can perform standalone client authentication and switch VLAN traffic locally even when it's disconnected to the WLC (Local Switching)

Question 51

What is a function of MAC address learning?

- A. It is disabled by default on all interfaces connected to trunks
- B. It increases security on the management VLAN
- C. It is enabled by default on all VLANs and interfaces
- D. It increases the potential for MAC address flooding

Answer: C

Question 52

A Cisco engineer at a new branch office is configuring a wireless network with access points that connect to a controller that is based at corporate headquarters. Wireless client traffic must terminate at the branch office and access-point survivability is required in the event of a WAN outage. Which access point mode must be selected?

- A. Lightweight with local switching disabled
- B. Local with AP fallback enabled
- C. OfficeExtend with high availability disabled
- D. FlexConnect with local switching enabled

Answer: D

Question 53

What is an advantage of using auto mode versus static mode for power allocation when an access point is connected to a PoE switch port?

- A. The default level is used for the access point
- B. It detects the device is a powered device
- C. All four pairs of the cable are used
- D. Power policing is enabled at the same time

Answer: B

Explanation

The switch supports these PoE modes:

auto – The switch automatically detects if the connected device requires power. If the switch discovers a powered device connected to the port and if the switch has enough power, it grants power, updates the power budget, turns on power to the port on a first-come, first-served basis, and updates the LEDs...

static – The switch pre-allocates power to the port (even when no powered device is connected) and guarantees that power will be available for the port.

Reference:

https://www.cisco.com/en/US/docs/switches/lan/catalyst3850/software/release/3.2_0_se/multibook/configuration_guide/b_consolidated_config_guide_3850_chapter_011010.html

Question 54

Refer to the exhibit.

```

R1# show ip route | begin gateway
Gateway of last resort is not set
  172.16.0.0/16 is variably subnetted, 5 subnets, 5 masks
O   172.16.2.128/25 [110/3184437] via 207.165.200.250, 00:00:24, Serial0/0/0
O   172.16.3.64/27 [110/3184437] via 207.165.200.250, 00:00:24, Serial0/0/0
O   172.16.3.128/28 [110/3184437] via 207.165.200.250, 00:00:24, Serial0/0/0
O   172.16.3.192/29 [110/3184437] via 207.165.200.250, 00:00:24, Serial0/0/0
O   172.16.4.0/23 [110/3184437] via 207.165.200.250, 00:00:24, Serial0/0/0
    207.165.200.0/24 is variably subnetted, 4 subnets, 2 masks
C   207.165.200.248/30 is directly connected, Serial0/0/0
L   207.165.200.249/32 is directly connected, Serial0/0/0
C   207.165.200.252/30 is directly connected, Serial0/0/1
L   207.165.200.253/32 is directly connected, Serial0/0/1

```

Drag and drop the learned prefixes from the left onto the subnet masks on the right.

172.16.3.128	255.255.254.0
172.16.3.64	255.255.255.128
172.16.2.128	255.255.255.224
172.16.3.192	255.255.255.240
172.16.4.0	255.255.255.248

Answer:

```

172.16.4.0 – 255.255.254.0
172.16.2.128 – 255.255.255.128
172.16.3.64 – 255.255.255.224
172.16.3.128 – 255.255.255.240
172.16.3.192 – 255.255.255.248

```

Question 55

Drag and drop the Ansible features from the left to the right. Not all features are used.

uses the YAML language	feature
executes modules via SSH by default	feature
pushes configurations to the client	feature
uses agents to manage hosts	feature
requires clients to pull configurations from the server	
operates without agents	

Answer:

uses the YAML language
 executes modules via SSH by default
 pushes configurations to the client
 operates without agents

Explanation

A comparison list of Ansible, Puppet and Chef automation tool:

Criteria	Ansible	Puppet	Chef
Configuration Language	YAML, Python	+ Puppet DSL + Embedded Ruby	Ruby DSL, JSON
Architecture	Agentless (Client only)	Both (Agentless & Agent Based)	Agent Based (Client-Server)
Deployment Method	Push Model	Pull Model	Pull Model
Configuration Style	Procedural	Declarative	Procedural
Files created before Operation	Playbook	Manifest	Recipe
Availability	Ansible Primary Instance	Puppet Master	Chef Master
How to manage devices	Any device (can become controller)	Puppet Master	Chef Master
Installation	easiest	medium	hard
Transport Mechanism	SSH/NETCONF	REST	REST
Port used	TCP port 22	TCP port 8140	TCP port 10002
Initial Release	2012	2005	2009

Question 56

Refer to the exhibit.

--Some output missing--

Routing Descriptor Blocks:

* directly connected, via Ethernet0/1

Route metric is 0, traffic share count is 1

CPE# ping 203.0.113.1

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 203.0.113.1, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/oax = 1/1/1 ms

CPE# show ip route

Gateway of last resort is 198.51.100.1 to network 0.0.0.0

B* 0.0.0.0/0 [20/0] via 198.51.100.1, 00:02:07

198.51.100.0/24 is variably subnetted, 2 subnets, 2 masks

C 198.51.100.0/30 is directly connected, Ethernet0/0

L 198.51.100.2/32 is directly connected, Ethernet0/0

203.0.113.0/24 is variably subnetted, 2 subnets, 2 masks

C 203.0.113.0/30 is directly connected, Ethernet0/1

L 203.0.113.2/32 is directly connected, Etheraet0/1

After configuring a new static route on the CPE, the engineer entered this series of commands to verify that the new configuration is operating normally. When is the static default route installed into the routing table?

- A. when the default route learned over external BGP becomes invalid
- B. when 203.0.113.1 is no longer reachable as a next hop
- C. when the default route learned over external BGP changes its next hop
- D. when a route to 203.0.113.1 is learned via BGP

Answer: A

Explanation

Maybe in the missing output at the top of the exhibit is about configuring a static default route (with command “ip route 0.0.0.0 0.0.0.0 ...”. But in the routing table of CPE router we only see a default route learned via BGP. We did not see the static default route because it was configured with a higher AD than that of the default route learned via BGP. Therefore only when the current default route becomes invalid then the default static route would be installed into the routing table.

Question 57

Refer to the exhibit.

The screenshot shows the Cisco IOS configuration interface for the Security section, specifically the 'Local Net Users > New' page. The left sidebar shows the navigation tree with 'Local Net Users' selected under the 'AAA' section. The main configuration area contains the following fields:

Field	Value
User Name	NA-User
Password	••••••••
Confirm Password	••••••••
Guest User	☒
Lifetime (seconds)	86400
Guest User Role	☐
WLAN Profile	Any WLAN ▼
Description	For NA WLAN Auth

Wireless LAN access must be set up to force all clients from the NA WLAN to authenticate against the local database. The WLAN is configured for local EAP authentication. The time that users access the network must not be limited. Which action completes this configuration?

- A. Check the Guest User Role check box
- B. Clear the Lifetime (seconds) value
- C. Set the Lifetime (seconds) value to 0
- D. Uncheck the Guest User check box

Answer: D

Explanation

The users created in the “Local Net Users” are stored in the local database. With the “Guest User” check box enabled, we have to set the time for that user. The “Lifetime” is the amount of time that the guest user account is to remain active. The valid range is 60 to 2,592,000 seconds (30 days) inclusive, and the default setting is 86,400 seconds.

If we don’t want to limit the time for that user, uncheck the “Guest User” check box.

===== New Questions (added on 7th-Nov-2022) =====

Question 58

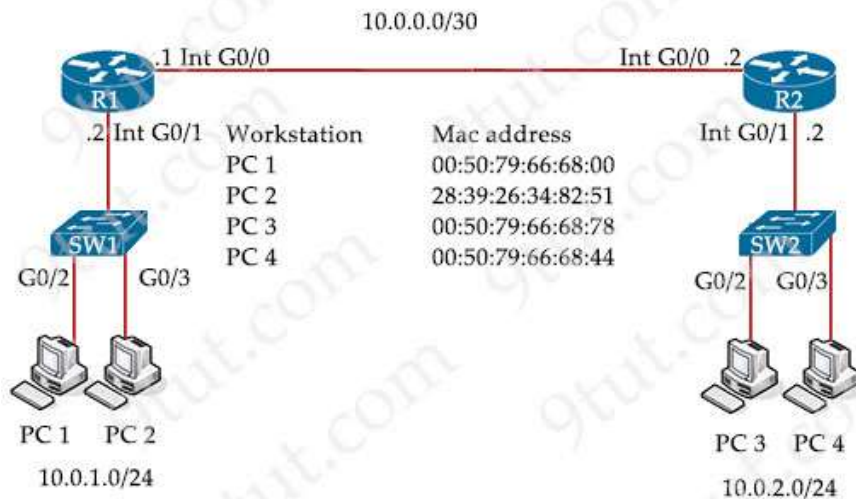
Which remote access protocol provides unsecured remote CLI access?

- A. Telnet
- B. SSH
- C. console
- D. Bash

Answer: A

Question 59

Refer to the exhibit.



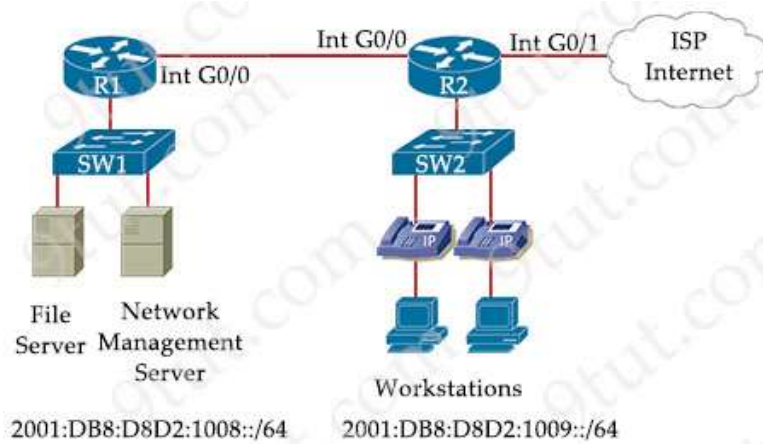
An engineer must configure the interface that connects to PC1 and secure it in a way that only PC1 is allowed to use the port. No VLAN tagging can be used except for a voice VLAN. Which command sequence must be entered to configure the switch?

- A. SW1(config-if)#switchport mode nonegotiate
SW1(config-if)#switchport port-security
SW1(config-if)#switchport port-security maximum 1
- B. SW1(config-if)#switchport mode access
SW1(config-if)#switchport port-security
SW1(config-if)#switchport port-security mac-address 0050.7966.6800
- C. SW1(config-if)#switchport mode dynamic desirable
SW1(config-if)#switchport port-security mac-address 0050.7966.6800
SW1 (config-if)#switchport port-security mac-address sticky
- D. SW1(config-if)#switchport mode dynamic auto
SW1(config-if)#switchport port-security
SW1(config-if)#switchport port-security violation restrict

Answer: B

Question 60

Refer to the exhibit.



A public IPv6 address must be configured for internet access. Which command must be configured on the R2 WAN interface to the service provider?

- A. ipv6 address fe80::260:3EFF:FE11:6770 link-local
- B. ipv6 address fe80: :/10
- C. ipv6 address 2001:db8:433:47:4620:ffff:ffff:ffff/64 anycast
- D. ipv6 address 2001:db8:123:45::4/64

Answer: D

Question 61

Drag and drop the device behaviors from the left onto the matching HSRP state on the right.

has heard from the neighbor device and is receiving hello packets	Learn
is forwarding packets	Active
is ready to forward packets if the device that is currently forwarding packets fails	Listen
is transmitting and receiving hello packets	Speak
is waiting to hear from the neighbor device	Standby

Answer:

- + is waiting to hear from the neighbor device: Learn
- + is forwarding packets: Active
- + has heard from the neighbor device and is receiving hello packets: Listen
- + is transmitting and receiving hello packets: Speak
- + is ready to forward packets if the device that is currently forwarding packets fails: Standby

Explanation

When in operation, HSRP devices are configured into one of many states:

Active – This is the state of the device that is actively forwarding traffic.

Init or Disabled – This is the state of a device that is not yet ready or able to participate in HSRP.

Learn – This is the state of a device that has not yet determined the virtual IP address and has not yet seen a hello message from an active device.

Listen – This is the state of a device that is receiving hello messages.

Speak – This is the state of a device that is sending and receiving hello messages.

Standby – This is the state of a device that is prepared to take over the traffic forwarding duties from the active device.

Reference: <https://www.pearsonitcertification.com/articles/article.aspx?p=2141271>

Question 62

Which two functions does a WLC perform in the lightweight access-point architecture that an AP performs independently in an autonomous architecture? (Choose two)

- A. handling the association, authentication, and roaming of wireless clients
- B. encrypting and decrypting traffic that uses the WAP protocol family
- C. preventing collisions between wireless clients on the same RF channel
- D. managing RF channels, including transmission power
- E. sending and processing beacon frames

Answer: A D

Explanation

This question asks what the WLC can do for the AP or which functions are moved from AP to WLC in lightweight access-point architecture.

The wireless clients encrypt and decrypt the traffic, not the AP -> Answer B is not correct.

The wireless clients also run CSMA/CA to prevent collisions between them, not the AP -> Answer C is not correct.

With wireless networking, we have real-time and management functions. The AP should handle real-time functions, but everything that is not delay-sensitive can do from a central location. We separate the following management and real-time functions of the AP:

+ Management functions:

- ++ Client authentication (-> Answer A is correct)
- ++ Security management
- ++ Association and reassociation (roaming)
- ++ Quality of Service (QoS)

- + Real-time functions:
- ++ Transmission of 802.11 frames
- ++ MAC management
- ++ Encryption

Since these functions are not real-time, we can move them to a central point, the WLC. We take away some of the intelligence of the AP, which is why we call them lightweight APs (LAP). We move this intelligence to the WLC.

Reference: <https://networklessons.com/cisco/ccna-200-301/cisco-wireless-network-architectures>

Radio Resource Management (RRM) allows the controller to dynamically control power and channel assignment of APs -> Answer D is correct.

Reference: <https://what-when-how.com/deploying-and-troubleshooting-cisco-wireless-lan-controllers/lightweight-compared-to-traditional-autonomous-aps-cisco-wireless-lan-controllers/>

Question 63

Refer to the exhibit.

Current Neighbor Relationship

Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.1.1	1	FULL/DR	00:00:33	192.168.1.1	GigabitEthernet0/0

Desired Neighbor Relationship

Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.1.1	0	FULL/ -	00:00:31	192.168.1.1	GigabitEthernet0/0

How must OSPF be configured on the GigabitEthernet0/0 interface of the neighbor device to achieve the desired neighbor relationship?

- A. Router(config)#interface GigabitEthernet 0/0
Router(config-if)#ip ospf cost 5
- B. Router(config)#interface GigabitEthernet 0/0
Router(config-if)#ip ospf 1 area 2
- C. Router(config)#interface GigabitEthernet 0/0
Router(config-if)#ip ospf network point-to-point
- D. Router(config)#interface GigabitEthernet 0/0
Router(config-if)#ip ospf priority 1

Answer: C

Explanation

We see the desired neighbor relationship does not have DR/BDR so we have to set the network type to “point-to-point”.

Question 64

What causes a port to be placed in the err-disabled state?

- A. latency
- B. nothing plugged into the port
- C. shutdown command issued on the port
- D. link flapping

Answer: D

Explanation

Link flap means that the interface continually goes up and down in a Cisco Switch. The interface is put into the errdisabled state if it flaps more than five times in 10 seconds. The common cause of link flap is a Layer 1 issue such as a bad cable, duplex mismatch, or bad Gigabit Interface Converter (GBIC) card. Look at the console messages or the messages that were sent to the syslog server that state the reason for the port shutdown.

```
13w0d: %PM-4-ERR_DISABLE: link-flap error detected on Fa0/28, putting Fa0/28 in err-disable state
```

Question 65

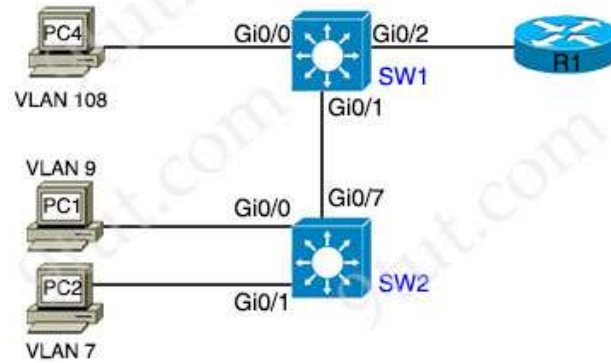
Which function forwards frames to ports that have a matching destination MAC address?

- A. frame pushing
- B. frame filtering
- C. frame flooding
- D. frame switching

Answer: D

Question 66

Refer to the exhibit.



The SW1 and SW2 Gi0/0 ports have been preconfigured. An engineer is given these requirements:

- + Allow all PCs to communicate with each other at Layer 3.
- + Configure untagged traffic to use VLAN 5.
- + Disable VLAN 1 from being used.

Which configuration set meets these requirements?

Option A <pre> SW1# interface Gi0/1 switchport mode trunk switchport trunk allowed vlan 5,7,9,108 interface Gi0/2 switchport mode trunk switchport trunk allowed vlan 7,9,108 SW2# interface Gi0/1 switchport mode trunk switchport trunk allowed vlan 7 interface Gi0/7 switchport mode trunk switchport trunk allowed vlan 5,7,9,108 </pre>	Option B <pre> SW1# interface Gi0/1 switchport mode trunk switchport trunk allowed vlan 5,7,9,108 switchport trunk native vlan 5 interface Gi0/2 switchport mode access switchport trunk allowed vlan 7,9,108 SW2# interface Gi0/1 switchport mode access no switchport access vlan 1 switchport access vlan 7 interface Gi0/7 switchport mode trunk switchport trunk allowed vlan 7,9,108 switchport trunk native vlan 5 </pre>
Option C	Option D

SW1# interface Gi0/1 switchport mode trunk switchport trunk allowed vlan 5,7,9,108 switchport trunk native vlan 5 interface Gi0/2 switchport mode trunk switchport trunk allowed vlan 5,7,9,108 SW2# interface Gi0/1 switchport mode access switchport access vlan 7 interface Gi0/7 switchport mode trunk switchport trunk allowed vlan 7,9,108	SW1# interface Gi0/1 switchport mode trunk switchport trunk allowed vlan 5,7,9,108 switchport trunk native vlan 5 interface Gi0/2 switchport mode trunk switchport trunk allowed vlan 5,7,9,108 SW2# interface Gi0/1 switchport mode access switchport access vlan 7 interface Gi0/7 switchport mode trunk switchport trunk allowed vlan 5,7,9,108 switchport trunk native vlan 5
--	--

- A. Option A
- B. Option B
- C. Option C
- D. Option D

Answer: D

Explanation

This question requires “Configure untagged traffic to use VLAN 5” so we must use the command “switchport trunk native vlan 5” on both Gi0/1 of Sw1 and Gi0/7 of Sw2 -> Only Option B and Option D are correct.

In order to allow all PCs to communicate with each other at Layer 3, interface Gi0/2 of SW1 must be configured as trunk, not access mode -> Only Option D is correct.

Question 67

Refer to the exhibit.

```
SW1#show ip interface brief
Interface      IP-Address    OK? Method Status Protocol
FastEthernet0/1 unassigned    YES manual down    down
```

```
SW1#show interface fa0/1 status
Port    Name    Status    Vlan    Duplex    Speed    Type
Fa0/1    notconnect 1        a-full  a-100    10/100BaseTX
```

What is the cause of the issue?

- A. STP
- B. shutdown command
- C. port security
- D. wrong cable type

Answer: D

Explanation

The “Status” of fa0/1 is “notconnect” so port security is not the issue or it would show the status of “err-disabled”.

“Shutdown” command is also not the issue as it would show “Administratively down” in the “Status” field of the “show ip interface brief” command.

With STP, even if the port is in “Block” (BLK) status but it is still in “up/up” state with the “show ip interface brief” command.

With wrong cable type, the port is in “down/down” state, same as there is no connected cable to the port.

Question 68

Refer to the exhibit.

```
Router1#show ip route
Gateway of last resort is 10.10.11.2 to network 0.0.0.0
  209.165.200.0/27 is subnetted, 1 subnets
B    209.165.200.224 [20/0] via 10.10.12.2,03:32:14
  209.165.201.0/27 is subnetted, 1 subnets
B    209.165.201.0 [20/0] via 10.10.12.2,02:26:53
  209.165.202.0/27 is subnetted, 1 subnets
B    209.165.202.128 [20/0] via 10.10.12.2,02:46:03
  10.0.0.0/8 is variably subnetted, 10 subnets, 4 masks
C    10.10.10.0/28 is directly connected, GigabitEthernet0/0
C    10.10.11.0/30 is directly connected, FastEthernet2/0
C    10.10.12.0/30 is directly connected, GigabitEthernet0/1
O    10.10.13.0/25 [110/2] via 10.10.10.1, 00:00:04, GigabitEthernet0/0
O    10.10.13.128/28 [110/2] via 10.10.10.1, 00:00:12, GigabitEthernet0/0
O    10.10.13.144/28 [110/2] via 10.10.10.1, 00:01:57, GigabitEthernet0/0
O    10.10.13.160/29 [110/2] via 10.10.10.1, 00:00:12, GigabitEthernet0/0
O    10.10.13.208/29 [110/2] via 10.10.10.1, 00:01:57, GigabitEthernet0/0
O    10.10.13.252/30 [110/2] via 10.10.10.1, 00:01:57, GigabitEthernet0/0
S*   0.0.0.0/0 [1/0] via 10.10.11.2
```

Drag and drop the subnet masks from the left onto the corresponding subnets on the right. Not all subnet masks are used.

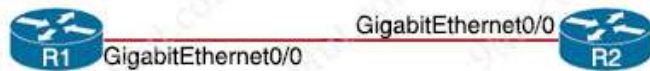
255.255.248.0	10.10.13.0
255.255.255.128	10.10.13.128
255.255.255.224	10.10.13.160
255.255.255.240	10.10.13.252
255.255.255.248	
255.255.255.252	

Answer:

10.10.13.0 – 255.255.255.128
 10.10.13.128 – 255.255.255.240
 10.10.13.160 – 255.255.255.248
 10.10.13.252 – 255.255.255.252

Question 69

Refer to the exhibit.



A network engineer must configure the link with these requirements:

- Consume as few IP addresses as possible.
- Leave at least two additional useable IP addresses for future growth.

Which set of configurations must be applied?

A. R1(config-if)#ip address 10.10.10.1 255.255.255.248
 R2(config-if)#ip address 10.10.10.4 255.255.255.248

B. R1(config-if)#ip address 10.10.10.1 255.255.255.240
 R2(config-if)#ip address 10.10.10.12 255.255.255.240

C. R1 (config-if)#ip address 10.10.10.1 255.255.255.252
R2(config-if)#ip address 10.10.10.2 255.255.255.252

D. R1 (config-if)#ip address 10.10.10.1 255.255.255.0
R2(config-if)#ip address 10.10.10.5 255.255.255.0

Answer: A

Question 70

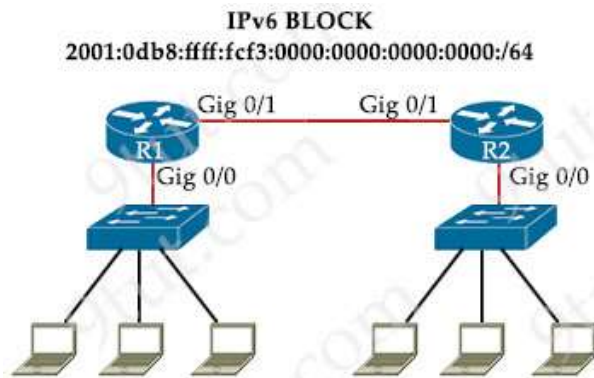
What is a function of Layer 3 switches?

- A. They route traffic between devices in different VLANs.
- B. They transmit broadcast traffic when operating in Layer 3 mode exclusively.
- C. They forward Ethernet frames between VLANs using only MAC addresses.
- D. They move frames between endpoints limited to IP addresses.

Answer: A

Question 71

Refer to the exhibit.



IPv6 is being implemented within the enterprise. The command **ipv6 unicast-routing** is configured. Interface Gig0/0 on R1 must be configured to provide a dynamic assignment using the assigned IPv6 block. Which command accomplishes this task?

- A. ipv6 address 2001:DB8:FFFF:FCF3::/64 eui-64
- B. ipv6 address 2001:DB8:FFFF:FCF3::/64 link-local
- C. ipv6 address 2001:0B8:FFFF:FCF3::1/64
- D. ipv6 address autoconfig 2001:DB8:FFFF:FCF2::/64

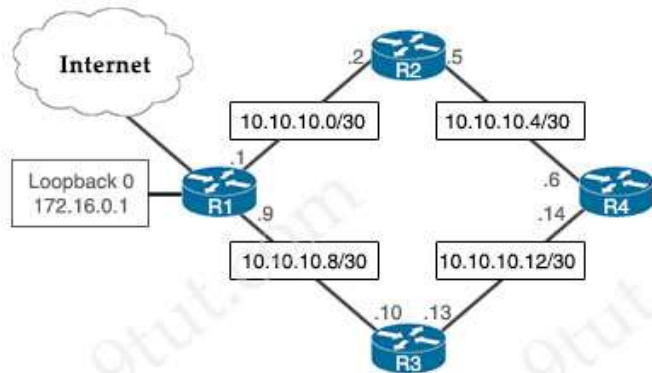
Answer: A

Explanation

Configure the IPv6 global unicast address on an interface using the **ipv6 address address/prefix-length [eui-64]** command.

Question 72

Refer to the exhibit.



```
R1#show run | include ntp
ntp master 7
ntp server 209.165.200.225
```

```
R2#show run | Include ntp
ntp server 172.16.0.1
```

```
R3#show run | include ntp
ntp master 6
ntp server 172.16.0.1
```

```
R4#show run | include ntp
ntp master 7
```

Which router or router group are NTP clients?

- A. R1, R2, and R3
- B. R1
- C. R2 and R3
- D R1, R3, and R4

Answer: A

Explanation

To configure the NTP client on a Cisco device we use the **ntp server** *{ip-address}* command in global configuration.

Question 73

A network engineer is replacing the switches that belong to a managed-services client with new Cisco Catalyst switches. The new switches will be configured for updated security standards, including replacing Telnet services with encrypted connections and doubling the modulus size from 1024. Which two commands must the engineer configure on the new switches? (Choose two)

- A. transport input ssh
- B. transport input all
- C. crypto key generate rsa general-keys modulus 1024
- D. crypto key generate rsa usage-keys
- E. crypto key generate rsa modulus 2048

Answer: A E

Explanation

The command “crypto key generate rsa modulus 2048” generate a 2048 bit RSA key pair (doubling the modulus size from 1024).

Question 74

Refer to the exhibit.

Gateway of last resort is not set
10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 10.1.1.0/30 is directly connected, GigabitEthernet0/0
L 10.1.1.2/32 is directly connected, GigabitEthernet0/0
S 192.168.0.0/20 [1/0] via 10.1.1.1
192.168.1.0/30 is subnetted, 1 subnets
S 192.168.1.0/30 [1/0] via 10.1.1.1
192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
S 192.168.2.0/28 [1/0] via 10.1.1.1
S 192.168.2.0/29 [1/0] via 10.1.1.1

An engineer is checking the routing table in the main router to identify the path to a server on the network. Which route does the router use to reach the server at 192.168.2.2?

- A. S 192.168.2.0/28 [1/0] via 10.1.1.1
- B. S 192.168.1.0/30 [1/0] via 10.1.1.1
- C. S 192.168.0.0/20 [1/0] via 10.1.1.1
- D. S 192.168.2.0/29 [1/0] via 10.1.1.1

Answer: D

Question 75

Refer to the exhibit.

```
{
  "myCar": {
    "name": "thunder",
    "wheels": ["good", "good", "pressureLow", "warning"],
    "gasLight": false
  },
  "oldCar": {
    "name": "sleepy",
    "wheels": ["pressureLow", "pressureLow", "pressureLow", "pressureLow"],
    "color": "rust"
    "gasLight": true
  },
  "newCar": {
    "name": "lightning",
    "wheels": ["pressureLow", "good", "pressureLow", "good"],
    "color": "blue"
    "gasLight": true
  }
}
```

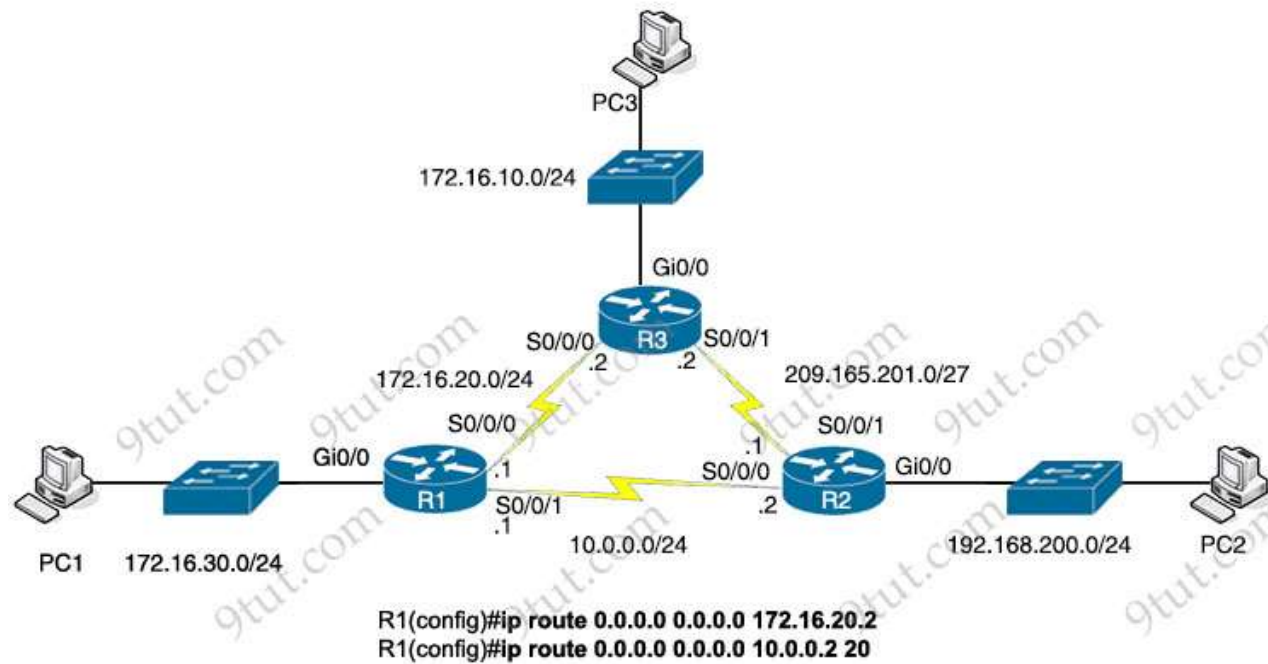
In which structure does the word “warning” directly reside?

- A. array
- B. object
- C. Boolean
- D. string

Answer: A

Question 76

Refer to the exhibit.



After applying this configuration to router R1, a network engineer is verifying the implementation. If all links are operating normally, and the engineer sends a series of packets from PC1 to PC3, how are the packets routed?

- A. They are routed to 172.16.20.2.
- B. They are distributed sent round robin to interfaces S0/0/0 and S0/0/1.
- C. They are routed to 192.168.100.2.
- D. They are routed to 10.0.0.2.

Answer: A

Question 77

A network administrator plans an update to the Wi-Fi networks in multiple branch offices. Each location is configured with an SSID called “Office”. The administrator wants every user who connects to the SSID at any location to have the same access level. What must be set the same on each network to meet the requirement?

- A. radio policy
- B. security policies
- C. NAS-ID configuration
- D. profile name

Answer: B

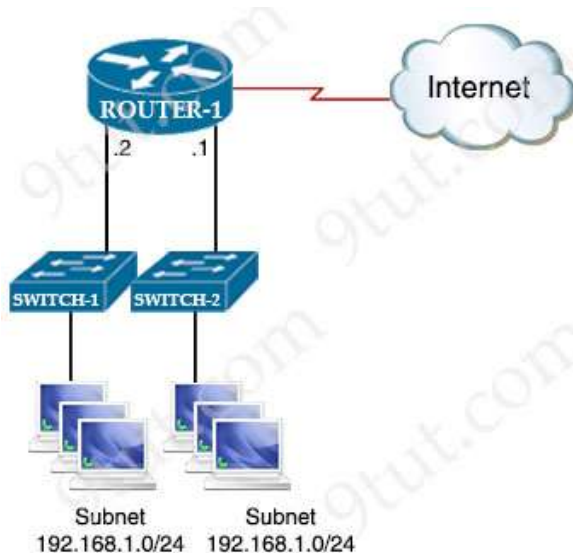
Explanation

Two identically named SSIDs with the same password will allow your device to connect to either, without having to add any extra networks on your devices. But make sure to configure them with the same security policies.

Note: Four WLAN security policies are available: Wired Equivalent Privacy (WEP), Wi-Fi Protected Access (WPA), WPA2, WLAN Authentication and Privacy Infrastructure (WAPI)

Question 78

Refer to the exhibit.



Which command set configures ROUTER-1 to allow Internet access for users on the 192.168.1.0/24 subnet while using 209.165.202.129 for Port Address Translation?

A. ip nat pool CCNA 192.168.0.0 192.168.1.255 netmask 255.255.255.0

```
access-list 10 permit 192.168.0.0 0.0.0.255
ip nat inside source list 10 pool CCNA overload
```

B. ip nat pool CCNA 209.165.202.129 209.165.202.129 netmask 255.255.255.255

```
access-list 10 permit 192.168.1.0 255.255.255.0
ip nat inside source list 10 pool CCNA overload
```

C. ip nat pool CCNA 192.168.0.0 192.168.1.255 netmask 255.255.255.0

```
access-list 10 permit 192.168.0.0 255.255.255.0
ip nat inside source list 10 pool CCNA overload
```

D. ip nat pool CCNA 209.165.202.129 209.165.202.129 netmask 255.255.255.255

```
access-list 10 permit 192.168.1.0 0.0.0.255
ip nat inside source list 10 pool CCNA overload
```

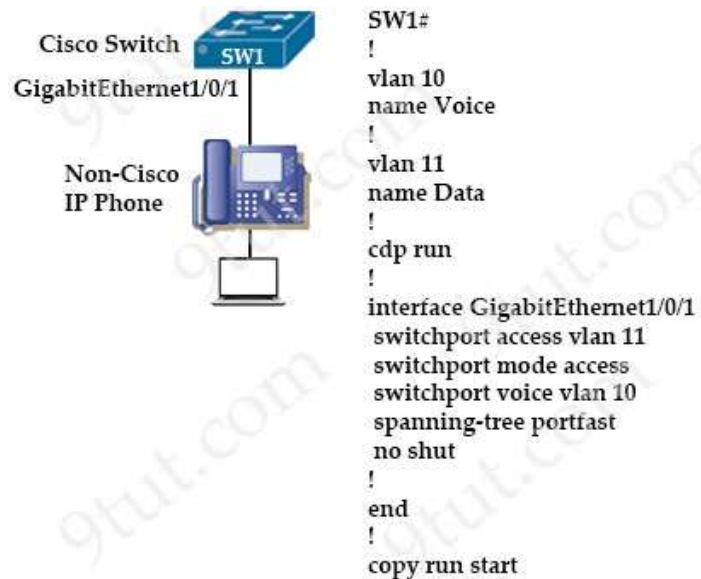
Answer: D

Explanation

By providing only one IP address in the pool CCNA, we are sure that IP address will be used for Internet with PAT.

Question 79

Refer to the exhibit.



A multivendor network exists and the company is implementing VoIP over the network for the first time. Which configuration is needed to implement the neighbor discovery protocol on the interface and allow it to remain off for the remaining interfaces?

- A. SW1(config)#no cdp enable
SW1 (config)#interface gigabitethernet1/0/1
SW1(config-if)#cdp run
- B. SW1(config)#lldp enable
SW1(config)#interface gigabitethernet1/0/1
SW1(config-if)#lldp run
- C. SW1(config)#lldp run
SW1 (config)#interface gigabitethernet1/0/1
SW1(config-if)#lldp enable

```
D. SW1(config)#no cdp run
SW1(config)#interface gigabitethernet1/0/1
SW1(config-if)#lldp transmit
SW1(config-if)#lldp receive
```

Answer: D

Explanation

There is no totally correct answer. LLDP is disabled by default so in fact we need one additional command to enable LLDP:

```
SW1(config)# lldp run
```

But this answer is still the best choice as we have to disable neighbor discovery protocol for the remaining interfaces with the “no cdp run” command (not “no cdp enable”).

Question 80

A network architect is considering whether to implement Cisco DNA Center to deploy devices on a new network. The organization is focused on reducing the time it currently takes to deploy devices in a traditional campus design. For which reason would Cisco DNA Center be more appropriate than traditional management options?

- A. Cisco DNA Center provides zero-touch provisioning to third-party devices.
- B. Cisco DNA Center supports deployment with a single pane of glass.
- C. Cisco DNA Center minimizes the level of syslog output when reporting on Cisco devices.
- D. Cisco DNA Center reduces the need for analytics on third-party access points and devices.

Answer: B

Explanation

DNA Center can be the “controller” or single pane of glass for deploying and managing on-premises Cisco networks including routing, switching, Software Defined Wireless Area Network (SD-WAN), and security devices like Cisco Identity Services Engine (ISE) and Firepower.

Question 81

How do TCP and UDP fit into a query-response model?

- A. TCP avoids using sequencing, and UDP avoids using acknowledgments.
- B. TCP uses error detection for packets, and UDP uses error recovery.
- C. TCP establishes a connection prior to sending data, and UDP sends immediately.
- D. TCP encourages out-of-order packet delivery, and UDP prevents re-ordering.

Answer: C

Explanation

TCP uses sequencing -> Answer A is not correct.

TCP uses error recovery while UDP uses error detection -> Answer B is not correct.

TCP does not encourage out-of-order packet delivery. When TCP packets are received out of order, they are buffered for later delivery to the application program rather than being discarded -> Answer D is not correct.

Question 82

Drag and drop the RF terms from the left onto the corresponding statements on the right.

absorption	measure of the minimum power required to decode a radio signal without excessive errors
receiver sensitivity	deviation from the propagation path that occurs when a signal encounters an obstacle
noise floor	reduction of energy in a signal as it travels away from the access point and encounters free space or obstacles
signal-to-noise ratio	measure of the total unwanted signals at the receiver
reflection	relative power of the desired radio signal to unwanted signals at the receiver

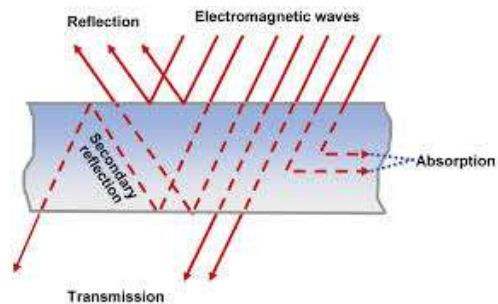
Answer:

- + measure of the minimum power required to decode a radio signal without excessive errors: **receiver sensitivity**
- + deviation from the propagation path that occurs when a signal encounters an obstacle: **reflection**
- + reduction of energy in a signal as it travels away from the access point and encounters free space or obstacles: **absorption**
- + measure of the total unwanted signals at the receiver: **noise floor**
- + relative power of the desired radio signal to unwanted signals at the receiver: **signal-to-noise ratio**

Explanation

The **noise floor** defines the amplitude of the sum of all noise sources and other unwanted signals within a system.

Sound **absorption** is the measure of the amount of energy removed from the sound wave as the wave passes through a given thickness of material.



Question 83

Which interface type enables an application running on a client to send data over an IP network to a server?

- A. southbound interface
- B. application programming interface
- C. northbound interface
- D. Representational State Transfer application programming interface

Answer: B

Explanation

In this question the answer “application programming interface” is better than the choice “Representational State Transfer application programming interface” as the latter is used for web services (HTTP/HTTPS) only.

Note: REpresentational State Transfer (REST) requires that a client make a request to the server in order to retrieve or modify data on the server. A request generally consists of:

- + an HTTP verb, which defines what kind of operation to perform
- + a header, which allows the client to pass along information about the request
- + a path to a resource
- + an optional message body containing data

Question 84

Refer to the exhibit.

```
access-list 10 permit 10.0.0.0 0.0.0.255
interface Serial0
ip access-list 10 in
```

A network administrator must permit traffic from the 10.10.0.0/24 subnet to the WAN on interface Serial0. What is the effect of the configuration as the administrator applies the command?

- A. The sourced traffic from IP range 10.0.0.0 – 10.0.0.255 is allowed on Serial0.
- B. The **permit** command fails and returns an error code.

- C. The router fails to apply the access list to the interface.
- D. The router accepts all incoming traffic to Serial0 with the last octet of the source IP set to 0.

Answer: C

Explanation

The last command is not correct. We must use the “ip access-group 10 in” command to apply this ACL to the interface.

Question 85

Refer to the exhibit.

Gateway of last resort is 172.16.2.2 to network 0.0.0.0

```
10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C   10.10.8.0/28 is directly connected, GigabitEthernet0/0/2
C   10.10.10.0/24 is directly connected, GigabitEthernet0/0/0
L   10.10.10.3/32 is directly connected, GigabitEthernet0/0/0
172.16.0.0/16 is variably subnetted, 3 subnets, 2 masks
S   172.16.1.33/32 is directly connected, GigabitEthernet0/0/1
C   172.16.2.0/23 is directly connected, GigabitEthernet0/0/1
L   172.16.2.1/32 is directly connected, GigabitEthernet0/0/1
S* 0.0.0.0/0 [1/0] via 172.16.2.2
```

A packet sourced from 10.10.10.1 is destined for 10.10.8.14. What is the subnet mask of the destination route?

- A. 255.255.254.0
- B. 255.255.255.240
- C. 255.255.255.248
- D. 255.255.255.252

Answer: B

Question 86

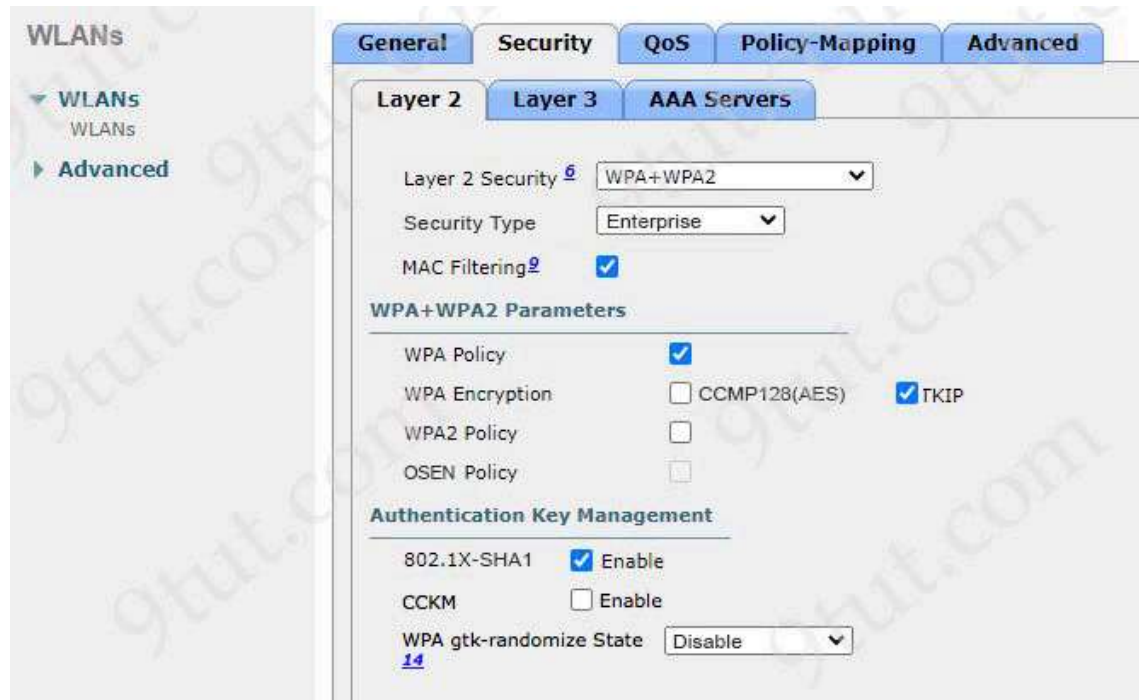
What are two reasons to implement IPv4 private addressing on a network? (Choose two)

- A. to expand the routing table on the router
- B. to facilitate renumbering when merging networks
- C. to enable internal applications to treat the private IPv4 addresses as unique
- D. to conserve global unique IPv4 addresses
- E. to provide protection from external denial-of-service attacks

Answer: D E

Question 87

Refer to the exhibit.



A guest WLAN must be created that prompts the client for a username and password on the local web page of the WLC. Which two actions must be performed on the Layer 2 tab before enabling the Authentication option on the Layer 3 tab? (Choose two)

- A. Uncheck the MAC Filtering option check box.
- B. Set the Security Type option to Personal.
- C. Change the WPA Encryption option from TKIP to CCMP128(AES).
- D. Set the Layer 2 Security option to None.
- E. Uncheck the WPA Policy option check box, and check the WPA2 Policy option check box.

Answer: A D

Explanation

We want to use Layer 3 authentication so we should disable Layer 2 authentication (by setting it to “None”).

Reference: <https://www.youtube.com/watch?v=6VIPjRdLSsY>

Also disable “MAC Filtering” as it is not supported with FlexConnect Local Authentication.

Question 88

Which script paradigm does Puppet use?

- A. manifests and modules
- B. strings and marionettes
- C. recipes and cookbooks
- D. playbooks and roles

Answer: A

Explanation

Puppet Master Components

Manifests

Manifest is the most important component in a Puppet Master so we will mention about it first. Manifest is just the file where the all Puppet scripts for configuring Puppet clients are written (in Ruby code). Manifest filenames use “.pp” (means Puppet policy) extension.

Based on the Facts received from Factor, Master compiles manifests into catalogs (which will be discussed later), then sends them to the client.

Module

Module also plays an important part in a Puppet Master. Module is a collection of manifests and other related data files organized in a predefined way to facilitate sharing and reusing. Modules tie manifests, templates, and files into a single unit.

Module = Manifests + Data (Templates, Files)

Modules have a specific directory path which is usually “/etc/puppet/manifests/”. They are useful for organizing our Puppet code, because they allow to split code into multiple manifests. It is considered best practice to use modules to organize all of our Puppet manifests.



Puppet Master

For more information about Puppet, please read our [Puppet tutorial](#).

Question 89

Which IPsec transport mode encrypts the IP header and the payload?

- A. pipe
- B. tunnel
- C. control
- D. transport

Answer: B

Explanation

In **tunnel** mode, the entire original IP packet is encapsulated to become the payload of a new IP packet. Additionally, a new IP header is added on top of the original IP packet.

The main difference in **transport** mode is that it retains the original IP header. In other words, payload data transmitted within the original IP packet is protected, but not the IP header.

Reference: <https://www.twingate.com/blog/ipsec-tunnel-mode/>

Question 90

What does WPA3 provide in wireless networking?

- A. increased security and requirement of a complex configuration
- B. backward compatibility with WPA and WPA2
- C. optional Protected Management Frame negotiation
- D. safeguards against brute force attacks with SAE

Answer: D

Explanation

WPA3 only backwards compatible with WPA2 but not WPA -> Answer B is not correct.

WPA3 increases security but not require a complex configuration -> Answer A is not correct.

Simultaneous Authentication of Equals (SAE): SAE provides a more secure, password-based authentication and key agreement mechanism even when passwords are not following complexity requirements. It protects from brute-force attacks and makes unwanted decrypting of sessions (during or after the session) a lot harder – just knowing the passphrase isn't enough to decrypt the session -> Answer D is correct.

Reference: <https://www.mist.com/wpa3-just-the-essentials-on-the-latest-in-wi-fi-security/>

Question 91

Refer to the exhibit. With which metric does router R1 learn the route to host 172.16.0.202?

```
R1#show ip route | begin gateway
```

```
Gateway of last resort is 209.165.200.246 to network 0.0.0.0  
S* 0.0.0.0/0 [1/0] via 209.165.200.246, Serial0/1/0
```

```
is directly connected, Serial0/1/0
  172.16.0.0/16 is variably subnetted, 3 subnets, 3 masks
S   172.16.0.0/24 [1/0] via 207.165.200.250, Serial0/0/0
O   172.16.0.128/25 [110/32445] via 207.165.200.254, 00:00:33, Serial0/0/1
D   172.16.0.192/29 [90/3184439] via 207.165.200.254, 00:00:33, Serial0/0/1
    207.165.200.0/24 is variably subnetted, 4 subnets, 2 masks
C   207.165.200.248/30 is directly connected, Serial0/0/0
L   207.165.200.249/32 is directly connected, Serial0/0/0
C   207.165.200.252/30 is directly connected, Serial0/0/1
L   207.165.200.253/32 is directly connected, Serial0/0/1
```

- A. 0
- B. 110
- C. 32445
- D. 3184439

Answer: C

Explanation

Both the line “O 172.16.0.128/25” and “S 172.16.0.0/24” cover the host 172.16.0.202 but with the “longest (prefix) match” rule the router will choose the first route.

Note: Network 172.16.0.192/29 does not cover host 172.16.0.202. For network 172.16.0.192/29:

Increment: 8
Network address: 172.16.0.192
Broadcast address: 172.16.0.199

Question 92

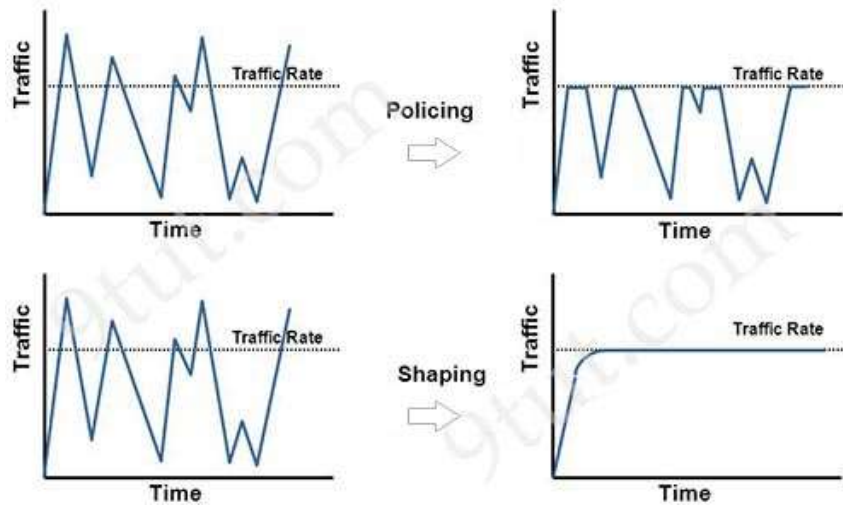
Which two actions are taken as the result of traffic policing? (Choose two)

- A. bursting
- B. fragmentation
- C. dropping
- D. remarking
- E. buffering

Answer: C D

Explanation

Policing: is used to control the rate of traffic flowing across an interface. During a bandwidth exceed (crossed the maximum configured rate), the excess traffic is generally dropped or remarked. The result of traffic policing is an output rate that appears as a saw-tooth with crests and troughs. Traffic policing can be applied to inbound and outbound interfaces. Unlike traffic shaping, QoS policing avoids delays due to queuing. Policing is configured in bytes.



QoS Traffic Shaping do not support Marking or **Remarking** again. This is done only with Traffic Policing.

Reference: <https://ipcisco.com/lesson/policing-and-shaping/>

Note: In policing, remarking the packets that exceed the CIR is an alternative to simply dropping them.

Question 93

Refer to the exhibit.

```

Gateway of last resort is 0.0.0.0 to network 0.0.0.0
10.0.0.0/8 is variably subnetted, 6 subnets, 5 masks
S    10.0.0.0/8 is directly connected, GigabitEthernet0/0
C    10.1.1.0/24 is directly connected, GigabitEthernet0/0
L    10.1.1.1/32 is directly connected, GigabitEthernet0/0
S    10.10.0.0/22 is directly connected, GigabitEthernet0/0
S    10.10.10.0/28 is directly connected, GigabitEthernet0/0
S    10.10.10.1/32 is directly connected, GigabitEthernet0/0
S*  0.0.0.0/0 is directly connected, GigabitEthernet0/0

```

Which IP route command created the best path for a packet destined for 10.10.10.3?

- A. ip route 10.0.0.0 255.0.0.0 g0/0
- B. ip route 10.10.10.1 255.255.255.255 g0/0
- C. ip route 10.10.10.0 255.255.255.240 g0/0
- D. ip route 10.10.0.0 255.255.252.0 g0/0

Answer: C

Explanation

The static routes here were configured as “directly connected static routes” (only exit interfaces were configured, not the next-hop IP address) so we see the line “is directly connected”.

All four answers of this question were configured and generated the output as shown in the exhibit. This question asks which command created the best path for packet destined to 10.10.10.3 so the correct one is answer C.

Question 94

A WLC sends alarms about a rogue AP, and the network administrator verifies that the alarms are caused by a legitimate autonomous AP. How must the alarms be stopped for the MAC address of the AP?

- A. Place the AP into manual containment.
- B. Remove the AP from WLC management.
- C. Manually remove the AP from Pending state.
- D. Set the AP Class Type to Friendly.

Answer: D

Explanation

If a rogue AP is classified as friendly, it means that the rogue AP exists in the vicinity, is a known AP, and need not be tracked. Therefore, all the rogue clients are either deleted or not tracked if they are associated with the friendly rogue AP.

Reference: https://content.cisco.com/chapter.sjs?uri=/searchable/chapter/content/en/us/td/docs/wireless/controller/7-5/configuration-guide/b_cg75/b_cg75_chapter_0111010.html.xml

Question 95

Which security method is used to prevent man-in-the-middle attack?

- A. authorization
- B. authentication
- C. anti-replay
- D. accounting

Answer: C

Explanation

The anti-replay protocol provides Internet Protocol (IP) packet-level security by making it impossible for a hacker to intercept message packets and insert changed packets into the data stream between a source computer and a destination computer.

How does the anti-replay protocol work?

The answer to preventing replay attacks is encrypting messages and including a key. IPsec provides anti-replay protection against attackers who could potentially intercept, duplicate or resend encrypted packets.

The mechanism uses a unidirectional security association to establish a secure connection between the source and destination nodes in the network and check whether a received

message is a replayed message. It works by assigning a monotonically increasing sequence number to each encrypted packet and then keeping track of the sequence numbers as packets arrive at the destination.

Reference: <https://www.techtarget.com/searchnetworking/definition/anti-replay-protocol>

[Comments \(29\)](#) Comments

Comment pages

[« Previous](#) [1](#) [2](#) 6259

1. Rictorres333

September 30th, 2022

More information: Using Packet Tracer Router 2911

A Works changing to : “aaa authentication login default local” instead of “aaa authentication login default radius”

B. Not works, error writing: login authentication default (authentication default command needed)

C. Works changing to: “login local” instead of “no login local”

D. Not works.

Not sure about syntax in this question. if anybody find it in you exam, tell us please.

2. JD

October 15th, 2022

@Rictorres333 and Q32:

Answer A works even with the “radius” line, but it has been misspelled. The proper radius command should be:

aaa authentication login default group radius

So in fact this is the best working answer here, but still the “aaa new-model” configuration is not a part of CCNA course agenda and not even in the CCNA 200-301 exam topics...

Complying to the ccna program the C answer would be the best if it contained “login local” instead of “no login local”. Maybe this question has been improperly dumped?

3. Anonymous

October 19th, 2022

Dear 9tut, Could you share file vce CCNA to me please ?

4. 9tut

October 20th, 2022

@Anonymous: We are sorry but we don't support VCE file.

5. lucky

October 26th, 2022

Some updates.

Just passed the exam. Is a 97Q exam with 2 labs and 20 new questions (new as in I did not encounter them in the big composite quiz). The coverage is more on the part 5-10 new questions. Perhaps it's time for 9tut to update the database.

6. Skotty

October 29th, 2022

Dears, Anyone knows if these questions are still valid? Also, i've heard multiple feedback that there are practical questions back in the exam. any idea about those practicals and what the questions might be?

7. Favzy

November 9th, 2022

Hello @9tut, please why am I not able to practice the newly added questions to part 10, from question 58?

8. 9tut

November 9th, 2022

@Favzy: We have just added the quiz from Q58 to 78. Currently we are too busy to create new quiz. We will create quiz for the rest tomorrow right after updating all the new questions on our site.

9. N84

November 9th, 2022

Q82

Answer:

+ deviation from the propagation path that occurs when a signal encounters an obstacle:
absorption

+ reduction of energy in a signal as it travels away from the access point and encounters free space or obstacles:
reflection

Absorption and reflection must be swapped in 9tut's answer. So the correct answer is:

+ deviation from the propagation path that occurs when a signal encounters an obstacle:
reflection

+ reduction of energy in a signal as it travels away from the access point and encounters free space or obstacles:
absorption

10. Favzy

November 9th, 2022

@9tut thank you. I await the new questions tomorrow. I'll be taking the exam this coming Saturday

11. JD

November 10th, 2022

The output of Q93 is correct... There surely can be “is directly connected” with “S” (static routes). They are called “directly connected static routes” and are created with “ip route” statement using only the exit interface without the next-hop address. Cheerz!

12. JD

November 10th, 2022

About Q93 explanation:

It surely can be “is directly conneted” with “S” (static route). They are called “directly connected static routes” and are established with “ip route” command using only the exit interface without forwarding router’s address. It’s all fine with this question.

13. 9tut

November 11th, 2022

@N84: Thank you for your detection, we updated RF Question!

@JD: Thank you for your information, we updated Q93 explanation!

@Favzy: Best wishes to you in your exam!

14. foreach

November 13th, 2022

Hello

@9tut thanks for the updates.

Can you check the following questions please ? I think there are some mistakes in the answers you suggest :

Question 42 :

According to you, the correct answers are :

C. SW(config-if)#switchport port-security mac-address sticky

E. SW(config-if)#switchport port-security violation restrict

But the question speaks about “MAC addresses (plural) must be learned dynamically”. I think :

1/ we can’t use sticky, or they won’t be learned after reboot.

2/ with MAC adresses in plural, it means we need to learn more than one MAC address.

So I think the correct answers should be :

B. SW(config-if)#switchport port-security maximum 2

E. SW(config-if)#switchport port-security violation restrict

Question 43 :

Your answers are :

C. is configured under the Layer 3 interface of a router on the client subnet

E. permits one IP helper command under an individual Layer 3 interface

I don’t understand why E. To me, it should be A. If we don’t use DHCP relay agent, we should have a DHCP server for each subnets on the network. So thanks to DHCP relay agents, we use less DHCP servers and that means answer A is correct.

Question 46 :

Drag and drop the steps in a standard DNS lookup operation from the left into the order on the right.

Answer:

Step 1: An endpoint submits a request for the IP address of a domain name

Step 2: The DNS submits a request to the domain DNS server

Step 3: The DNS submits a request to a root DNS server

Step 4: The DNS receives a reply from the domain DNS server

Step 5: The DNS responds to the endpoint

=> I think you mixed steps 2 and 3. According to your explanation, step 2 should be the request to the root DNS server and step 3, the request to the domain DNS server.

Question 79 :

LLDP is not activated globally according to the exhibit. And using “lldp transmit/receive” in an interface isn’t enough to activate LLDP. The least bad answer here seems to be C. It activates LLDP but it doesn’t deactivate it for the remaining interfaces.

Question 83

Which interface type enables an application running on a client to send data over an IP network to a server?

A. southbound interface

B. application programming interface

C. northbound interface

D. Representational State Transfer application programming interface

=> You’re marking D as the correct answer. But REST APIs are for web-based applications and the question doesn’t state about a web app. Shouldn’t the correct answer be B, API ?

Question 104 :

Correct answer should be A (hello and dead timers) and not D (just dead timer).

Thanks !

15. Rachide

November 13th, 2022

Hi 9tut, for question 104, I think the correct answer is A because for R1 we have: Hello 15, Dead 20, Wait 20, Retransmit 5 and R2 we have: Hello 10, Dead 40 , Wait 40, Retransmit 5

Option A

```
R2(config)#interface g0/0/0
```

```
R2(config-if)#ip ospf hello-interval 15
```

R2(config-if)#ip ospf dead-interval 20

Option D

R2(config)#interface g0/0/0

R2(config-if)#ip ospf dead-interval 20

16. 9tut

November 14th, 2022

@foreach, @Rachide: Thank you for your information, we updated them!

17. tom

November 19th, 2022

What happened to q97 to 137 in part 10, they are now missing?

18. 9tut

November 20th, 2022

@tom: This post was too long so we moved them to <https://www.9tut.com/ccna-new-questions-part-11>

19. Pobby

July 21st, 2023

Which encryption method is used by WPA3?

A. TKIP

B. SAE

C. PSK

D. AES >> Correct

Answer should be AES Under GCMP Mode as it asks for encryption not Authnetication

20. Pobby

July 21st, 2023

@9tut

21. Viral

September 1st, 2023

@9tut

Question 25:

The correct answer is AE

SAE is authentication method, not encryption

Could you please update the answer?

22. Anonymous

February 17th, 2024

@ 9tut & Anonymous

“Regarding “Question 11”:

The answer is correct because IP 10.1.2.126 is in Subnet 10.1.2.112/28.

Network: 10.1.2.112/28 (Class A)

Netmask: 255.255.255.240

Host Address Range: 10.1.2.113-10.1.2.126 (14 hosts)

Broadcast: 10.1.2.127”

options A, B , C, D should be:

A. 10.145.20.126 NOT 10.165.20.126

B. 10.145.20.146 NOT 10.165.20.146

C. 10.165.20.166 correct answer (10.1.2.112/28 more specific)

D. 10.145.20.226 NOT 10.165.20.226

(for 10.1.2.112 /28 (more specific) first usable is .2.113 and last usable .2.126)

(for 10.1.2.0 /25 first usable is .2.1 and last usable .2.126)

Thank you

23. Zkin

December 1st, 2024

@9tut Which two IPv6 addresses are used to provide connectivity between two routers on a shared link? (Choose two)

FF02 and 2001 are the correct answers. FF02 is a neighbor solicited message. The keyword is “shared link” although ::ff is a valid answer but its not specifically used for router-to-router connectivity on a “shared link”

24. 9tut

December 1st, 2024

@Zkin: We checked this question again but still believe our answers are correct. If answer “FF02...” is correct then answer “ff06:bb43:cc13:dd16:1bb:ff14:7545:234d” is correct too as they both are multicast addresses.

25. Zkin

December 7th, 2024

@9tut Another clarification is the wi-fi terms:

+ entire wireless cell of an access point and the linkage to the wired network: Distribution mode

+ Wi-Fi option based around one or more access points: Infrastructure mode

Can you please double check?

26. 9tut

December 8th, 2024

@Zkin: We believe our answers for this question are correct.

27. gulyev

February 23rd, 2025

Hello, I think the correct answer is C at question 69. Pls explain why you choose the A ?

The goal is to consume as few IP addresses as possible while leaving at least two additional usable addresses.

/30 (255.255.255.252) subnet provides 4 total addresses, with 2 usable IPs for router interfaces, 1 network address, and 1 broadcast address. This is ideal for point-to-point links, as it meets both conditions:

Other Option:

Option A (/29 – 255.255.255.248) → Uses 8 total IPs, but only 6 are usable. This wastes IPs for a simple point-to-point link. ❌

Thus, Option C (/30 subnet) is the best choice for a point-to-point connection with minimal IP waste.

28. arklst1976

May 1st, 2025

Q42

I really don't think Q42 answer is B & E

E is correct

I believe C is also correct... "sticky" MAC addresses are dynamically learned and then saved in the switch's running configuration

There's no mention of what happens after a reboot, nor is their mention of a maximum MACs allowed on the port.

I don't like to answer Qus based on info that isn't there.

29. scooby-does

July 20th, 2025

All routers in the network are configured. R2 must be the DR. After the engineer connected the devices, R1 was elected as the DR. Which command sequence must be configure on R2 to be elected as the DR in the network?

The answer should be B ie DR election based on router ID. No priorities are mentioned in the question, so we cannot assume setting priority to 100 will get the desired router elected as DR. WE can however presume that all priorities are equal and do a DR election based on router Id

Comment pages

[« Previous](#) 1 2 6259

Add a Comment

	Name

Submit Comment

[Subscribe to comments feed](#)

[CCNA – New Questions Part 10 Question 1 to 20 OSPF Configuration Sim](#)

Premium Member Zone

Welcome [Mentor Raphael!](#)

- [Welcome Premium Member](#)
- [CCNA – New Questions Part 5](#)
- [CCNA – New Questions Part 6](#)
- [CCNA – New Questions Part 7](#)
- [CCNA – New Questions Part 8](#)
- [CCNA – New Questions Part 9](#)
- **[CCNA – New Questions Part 10](#)**
- [CCNA – New Questions Part 11](#)
- [CCNA – New Questions Part 12](#)
- [CCNA – New Questions Part 13](#)
- [CCNA – New Questions Part 14](#)
- [CCNA – New Questions Part 15](#)
- [Composite Quizzes](#)
- ===== Lab Sims for CCNA (please learn all of them along with "New CCNA v1.1 Lab Sims" below) =====
- [IP Services Sim](#)
- [IP Services Sim Version 2](#)
- [Static Routing Configuration Sim](#)
- [OSPF Configuration Sim](#)
- [LACP Configuration Sim](#)
- [Voice VLAN Configuration Sim](#)
- [VLAN and Trunking Configuration Sim](#)
- [Named Access-list & Port Security Sim](#)
- [Named Access-list & DHCP Snooping Sim](#)
- [VLAN and CDP Sim](#)
- [IPv4 and IPv6 Deployment Sim](#)
- [Static Routing Configuration Sim 3](#)
- [VLAN & LLDP Sim](#)
- [IPv6 Static Routing Sim](#)
- [Voice VLAN Configuration Sim 2](#)
- [Named Access-list & DHCP Snooping Sim 2](#)
- [Voice VLAN & LLDP Sim](#)
- [ISL Trunking and LACP Sim](#)
- ===== New CCNA v1.1 Lab Sims =====
- [Allowed, Native VLAN & LACP Sim](#)
- [Static Routing Configuration Sim 4](#)
- [IPv4 and IPv6 Connectivity Sim](#)
- [VLAN & CDP Sim 2](#)
- [IPv4 and IPv6 Assignment Sim](#)
- [VLAN CDP & LLDP Sim](#)
- [OSPF Configuration Sim 2](#)
- [OSPF Configuration Sim 3](#)
- [Trunking & LACP Sim](#)
- [VLAN Configuration Sim](#)

- [Allowed, Native VLAN & LACP Sim 2](#)
- [802.1Q Trunking & LACP](#)
- [IPv4 and IPv6 Assignment Sim 2](#)
- [802.1Q Trunking, Native VLAN & LACP](#)
- [Static Routing Configuration Sim 5](#)
- [Static Routing Configuration Sim 6](#)
- === Below Sims are old and retired so you do not need to learn them ===
- [Static Routing Configuration Sim 2](#)

- [Logout](#)

CCNA Training

- [CCNA Knowledge](#)
- [CCNA Tutorials, Practice Labs & Lab Challenges](#)
- [Spine-Leaf Architecture Tutorial](#)
- [Generative AI vs Predictive AI Tutorial](#)
- [REST API Tutorial](#)
- [Network Time Protocol \(NTP\) Tutorial](#)
- [DHCP Snooping Tutorial](#)
- [What is SDN?](#)
- [How a router makes a forwarding decision](#)
- [Configure SSH for Remote Access on Cisco Router](#)
- [Key Security Concepts Tutorial](#)
- [Forwarding per-hop behavior \(PHB\) for QoS Tutorial](#)
- [Port Security Tutorial](#)
- [Puppet Tutorial](#)
- [Chef Tutorial](#)
- [Ansible Tutorial](#)
- [JSON Tutorial](#)
- [Layer 2 Threats and Security Features](#)
- [AAA TACACS+ and RADIUS Tutorial](#)
- [STP Root Port Election Tutorial](#)
- [TCP and UDP Tutorial](#)
- [WAN Tutorial](#)
- [DHCP Tutorial](#)
- [Simple Network Management Protocol SNMP Tutorial](#)
- [Syslog Tutorial](#)
- [EtherChannel Tutorial](#)
- [Hot Standby Router Protocol HSRP Tutorial](#)
- [InterVLAN Routing Tutorial](#)
- [Cisco Command Line Interface CLI](#)
- [Cisco Router Boot Sequence Tutorial](#)
- [OSI Model Tutorial](#)
- [Subnetting Tutorial – Subnetting Made Easy](#)
- [Frame Relay Tutorial](#)
- [Wireless Tutorial](#)

- [Virtual Local Area Network VLAN Tutorial](#)
- [VLAN Trunking Protocol VTP Tutorial](#)
- [IPv6 Tutorial](#)
- [Rapid Spanning Tree Protocol RSTP Tutorial](#)
- [Spanning Tree Protocol STP Tutorial](#)
- [Network Address Translation NAT Tutorial](#)
- [Access List Tutorial](#)
- [RIP Tutorial](#)
- [EIGRP Tutorial](#)
- [OSPF Tutorial](#)
- [Practice CCNA GNS3 Labs](#)
- [CCNA Lab Challenges](#)

CCNA 200-301

- [CCNA FAQs & Tips](#)
- [Basic Questions](#)
- [Topology Architecture Questions](#)
- [Cloud & Virtualization Questions](#)
- [CDP & LLDP Questions](#)
- [Switch Questions](#)
- [VLAN & Trunking Questions](#)
- [VLAN & Trunking Questions 2](#)
- [STP & VTP Questions](#)
- [EtherChannel Questions](#)
- [TCP & UDP Questions](#)
- [IP Address & Subnetting Questions](#)
- [IP Routing Questions](#)
- [IP Routing Questions 2](#)
- [OSPF Questions](#)
- [OSPF Questions 2](#)
- [EIGRP Questions](#)
- [NAT Questions](#)
- [NTP Questions](#)
- [Syslog Questions](#)
- [HSRP Questions](#)
- [Access-list Questions](#)
- [AAA Questions](#)
- [Security Questions](#)
- [Security Questions 2](#)
- [DAI Questions](#)
- [IPv6 Questions](#)
- [DNS Questions](#)
- [QoS Questions](#)
- [Port Security Questions](#)
- [Wireless Questions](#)

- [Wireless Questions 2](#)
- [SDN Questions](#)
- [DNA Center Questions](#)
- [Drag Drop Questions](#)
- [Drag Drop Questions 2](#)
- [Drag Drop Questions 3](#)
- [VPN Questions](#)
- [DHCP Questions](#)
- [Automation Questions](#)
- [Miscellaneous Questions](#)
- [Share your CCNA Experience](#)

Network Resources

- [Free Router Simulators](#)
- [CCNA Website](#)
- [ENCOR Website](#)
- [ENSDWI Website](#)
- [ENARSI Website](#)
- [DevNet Website](#)
- [CCIE R&S Website](#)
- [Security Website](#)
- [Wireless Website](#)
- [Design Website](#)
- [Data Center Website](#)
- [Service Provider Website](#)
- [Collaboration Website](#)

[Top](#)



Copyright © 2014 - 2025 CCNA Training
[Privacy Policy](#) | [Disclaimer](#) | [Contact Us](#)